# PHYSICS

1) A bar magnet has coercivity $4 \times 10^3$ $Am^{-1}$. It is desired to demagnetise it by inserting it inside a solenoid 12 cm long and having 60 turns. The current that should be sent through the solenoid :-

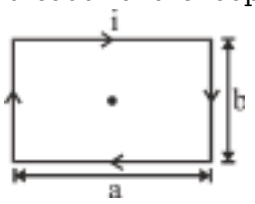
(1) 2 A

(2) 4 A

(3) 6 A

(4) 8 A

2) Adjoining figure shows a rectangular loop of conductor carrying a current i. The length and breadth of the loop are respectively a and b. The magnetic field at the centre of loop is :-



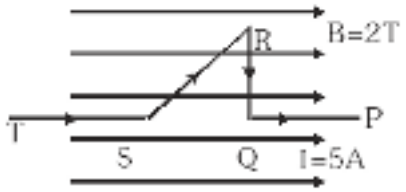

(1) $\frac{\mu_0 i(a+b)}{2\pi\sqrt{a^2+b^2}}$

(2) $\frac{\mu_0 iab}{2\pi\sqrt{a^2+b^2}}$

(3) $\frac{\mu_0 i(a+b)}{\pi\sqrt{a^2+b^2}}$

(4) $\frac{2\mu_0 i\sqrt{a^2+b^2}}{\pi ab}$

3) A wire PQRST carrying current I = 5A is placed in uniform magnetic field B = 2T as shown in fig. If the length of part QR = 4 cm and SR = 6 cm then the magnetic force on SR edge of the wire is :-



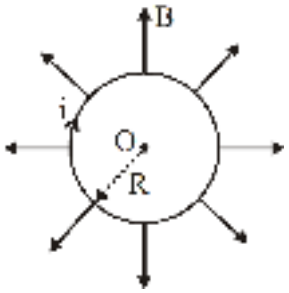

(1) 0.4 N

(2) 0.6 N

(3) zero

(4) 6 N

4) A current carrying circular loop of radius R is placed in radial magnetic field B according to figure




then the magnetic force on loop is :-

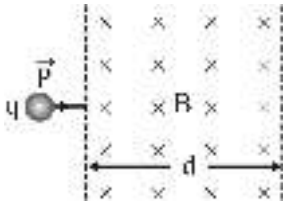
(1) Bi(2πR)

(2) Bi(πR)

(3) Bi(4πR)

(4) Zero

5) A particle with charge q, moving with a momentum p, enters a uniform magnetic field normally. The magnetic field has magnitude B and is confined to a region of width d. where $d < \frac{p}{Bq}$. The particle is deflected by an angle θ in crossing the field then -




(1) $\sin\theta = \frac{Bqd}{p}$

(2) $\sin\theta = \frac{p}{Bqd}$

(3) $\sin\theta = \frac{Bp}{qd}$

(4) $\sin\theta = \frac{pd}{Bq}$

6) A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b respectively. When a current I passes through the coil, the magnetic field at the centre is:

(1) $\mu_0 NI/b$

(2) $2\mu_0 NI/a$

(3) $\frac{\mu_0 NI}{2(b-a)}\ell n\frac{b}{a}$

(4) $\frac{\mu_0 NI}{(b-a)}\ell n\frac{b}{a}$

7) Mean radius of a toroid is 10 cm and number of turns are 500. If current flowing through it is 0.1 ampere then value of magnetic induction (in tesla) for toroid :-

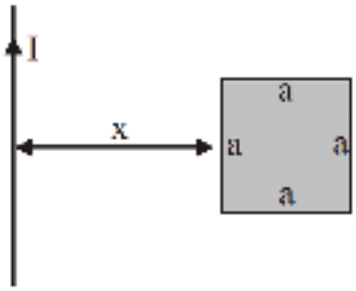
(1) $10^{-2}$

(2) $10^{-5}$

(3) $10^{-3}$

(4) $10^{-4}$

8) The mutual inductance between a long straight wire and a square loop of side a as shown in figure will be :-



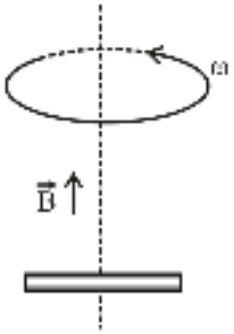

(1) $\frac{\mu_0 a}{2\pi}\ell n\left(\frac{x+a}{x}\right)$

(2) $\frac{\mu_0 a}{4\pi}\ell n\left(\frac{x+a}{x}\right)$

(3) $\frac{\mu_0 a}{\pi}\ell n\left(\frac{x+a}{x}\right)$

(4) $\frac{\mu_0 a}{2\pi}\ell n\left(\frac{a}{x}\right)$

9) A conducting rod of length 2ℓ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The e.m.f. induced between two ends of the rod is :-



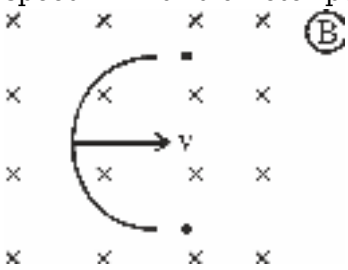

(1) $B\omega\ell^2$

(2) $\frac{1}{2}B\omega\ell^2$

(3) $\frac{1}{8}B\omega\ell^2$

(4) Zero

10) A straight wire of length L is bent into semicircle. It is moved in a uniform magnetic field with speed 'v' with diameter perpendicular to the field. The induced emf between the ends of the wire is :-



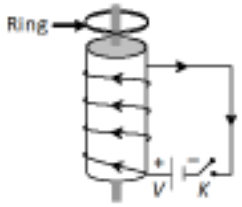

(1) BLv

(2) 2 BLv

(3) 2π BLv

(4) $\frac{2BvL}{\pi}$

11) A conducting ring is placed around the core of an electromagnet as shown in fig. When key K is pressed, the ring




(1) Remain stationary

(2) Is attracted towards the electromagnet

(3) Jumps out of the core

(4) None of the above

12) The number of turns in the coil of an ac generator is 5000 and the area of the coil is 0.25 $m^2$. The coil is rotated at the rate of 100 cycles/sec in a magnetic field of 0.2 $Wb/m^2$. The peak value of the emf generated is nearly :

(1) 786 kV

(2) 440 kV

(3) 220 kV

(4) 157.1 kV

13) Power dissipated in an LCR series circuit connected to an a.c. source of emf 2ε is :-

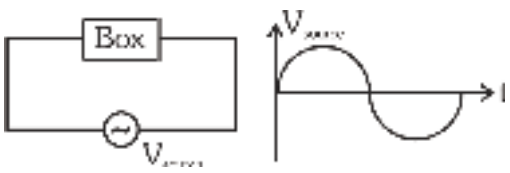
(1) $\varepsilon^2 R/\sqrt{R^2+\left(L\omega-\frac{1}{C\omega}\right)^2}$

(2) $4\varepsilon^2 R/\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]$

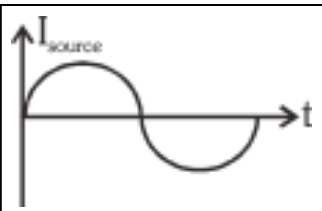
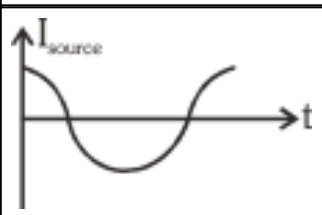
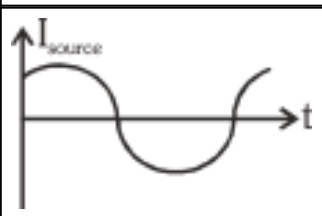
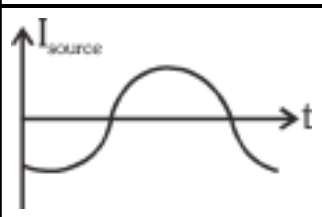
(3) $\varepsilon^2\sqrt{\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}/R$

(4) $\frac{\varepsilon^2\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}{R}$

14) Box may have any series combination of L, C and R.




| | Column-I | | Column-II |
|---|---|---|---|

| | | | |
|---|---|---|---|
| (A) |  | (P) | Box may contain LCR |
| (B) |  | (Q) | Box may contain only LR |
| (C) |  | (R) | Power factor of box zero |
| (D) |  | (S) | Box may contain only CR |
| | | (T) | State of resonance |

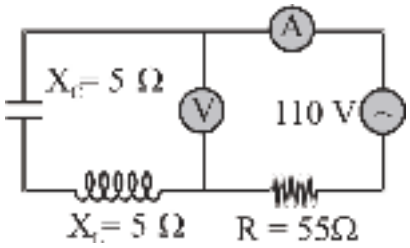
(1) A → P, T; B → R; C → P, S; D → P, Q

(2) A → P, R; B → S; C → T; D → P, Q

(3) A → P, S; B → R; C → T; D → P, Q

(4) A → P, T; B → T; C → R; D → P, Q

15) The reading of ammeter in the circuit shown will be :-




(1) 2 A

(2) 2.4 A

(3) Zero

(4) 1.7 A

16) A parallel plate capacitor with plate area A and separation between the plates d, is charged by a constant current I. Consider a plane surface of area A/2 parallel to the plates and drawn between the plates. The displacement current through the area is

(1) I

(2) $\frac{I}{2}$

(3) $\frac{I}{4}$

(4) $\frac{I}{8}$

17) Which of the following has/have zero average value in a plane electromagnetic waves?

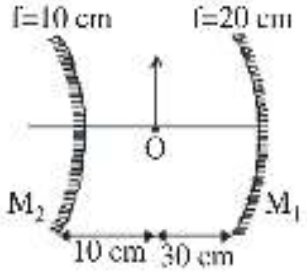
(1) Both magnetic and electric fields

(2) Electric field only

(3) Magnetic field only

(4) None of these

18) In figure, find the total magnification after two successive reflections first on $M_1$ and then on $M_2$.




(1) –2

(2) +1

(3) + 2

(4) – 1

19) A rod of length 5 cm is placed along the principal axis in front of a concave mirror of focal length 10 cm, distance of near end of rod from mirror is 10 cm. The length of image of rod is :-

(1) 2 cm

(2) ∞

(3) 1 cm

(4) Image will not form

20) A double convex lens of focal length 6 cm is made of glass of refractive index 1.5. The radius of curvature of one surface is double that of other surface. The value of small radius of curvature is :

(1) 6 cm

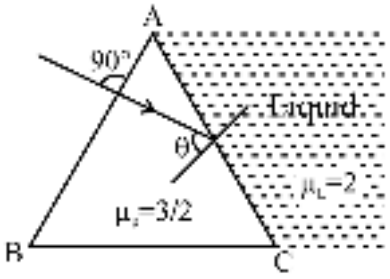
(2) 4.5 cm

(3) 9 cm

(4) 4 cm

21) A ray of light incident normally on surface AB of prism. What should be value of θ as shown in diagram such that it returns back to prism again :-

[If R. I. of prism $\mu_p = \frac{3}{2}$, R. I. of liquid $\mu_L = 2$]



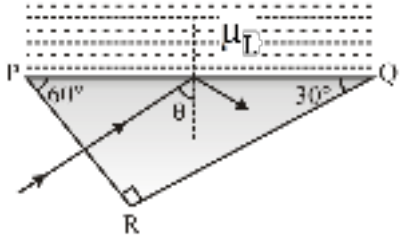

(1) $\theta \geqslant \sin^{-1}\left(\frac{6}{4}\right)$

(2) $\theta \geqslant \sin^{-1}\left(\frac{4}{6}\right)$

(3) θ = 90°

(4) Not possible

22) PQR is a right angled prism with other angles as 60° and 30°. Refractive index of prism is 1.5. Surface PQ has a thin layer of liquid over it. Light falls normally on the face PR. For total internal reflection, refractive index of liquid is :-



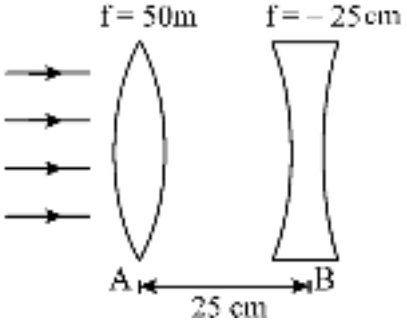

(1) 1.4

(2) 1.2

(3) 1.5

(4) 1.6

23) The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly towards lens A. The beam emerging from lens B is-




(1) Initially parallel and then diverging

(2) Always diverging

(3) Initially converging and finally parallel

(4) Always parallel

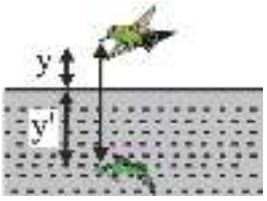
24) A thin glass lens with focal length f is to be used as a magnifying glass. Which of the following statements regarding the situation is true?

(1) A converging lens may be used, and the object be placed at a distance greater than 2f from the lens.

(2) A diverging lens may be used, and the object be placed between f and 2f from the lens.

(3) A converging lens may be used, and the object be placed at a distance less than f from the lens.

(4) A diverging lens may be used, and the object be placed at any point other than the focal point.

25) A fish rising vertically up, towards the surface of water with speed 3 $ms^{-1}$ observes a bird diving vertically down towards it with speed 9 $ms^{-1}$. The actual velocity of bird and velocity of fish as seen by bird will be :- 


(1) 4.5 $ms^{-1}$, 6.75 $ms^{-1}$

(2) 4 $ms^{-1}$, 6.75 $ms^{-1}$

(3) 4.5 $ms^{-1}$, 6 $ms^{-1}$

(4) 4 $ms^{-1}$, 6 $ms^{-1}$

26) For an object placed at a distance 2.4 m from a lens, a sharp focused image is observed on a screen placed at a distance 12 cm from the lens. A glass plate of refractive index 1.5 and thickness 1 cm is introduced between lens and screen such that the glass plate plane, faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screen?

(1) 0.8 m

(2) 3.2 m

(3) 1.2 m

(4) 5.6 m

27) Two identical coherent sources produces a zero order bright fringe on a screen. If β is fringe width then find the minimum distance between two points on either side of centre where intensity is half of maximum intensity

(1) $\frac{\beta}{2}$

(2) $\frac{\beta}{4}$

(3) $\frac{\beta}{3}$

(4) $\frac{\beta}{6}$

28) Two monochromatic light waves of amplitude 3A and 2A interfering at a point have a phase difference of 60°. The square of resultant amplitude at that point will be :-

(1) $19A^2$

(2) $13A^2$

(3) $7A^2$

(4) $5A^2$

29) If in Fraunhofer diffraction due to a single slit, the slit width is increased, the width of the central maximum will -

(1) Increase

(2) Decrease

(3) Not change

(4) Change depending upon the wavelength of light used

30) Two polaroids $P_1$ and $P_2$ are placed with their axis parallel to each other. Unpolarised light $I_0$ is incident on $P_1$. A third polaroid $P_3$ is kept in between $P_1$ and $P_2$ such that its axis makes an angle 30° with that of $P_1$. The intensity of transmitted light through $P_2$ is :-

(1) $\frac{I_0}{8}$

(2) $\frac{3I_0}{32}$

(3) $\frac{9I_0}{32}$

(4) $\frac{I_0}{32}$

31) A stationary nucleus ruptures into two nuclear parts which have their velocity ratio equal to 2 : 1. What will be the ratio of their nuclear size (nuclear radius):-

(1) $2^{1/3} : 1$

(2) $1 : 2^{1/3}$

(3) $3^{1/2} : 1$

(4) $1 : 3^{1/2}$

32) A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector ?

(1) 2 photon of energy 10.2 eV

(2) 2 photon of energy 1.4 eV

(3) One photon of energy 10.2 eV and an electron of kinetic energy 1.4 eV

(4) One photon of energy 10.2 eV and another photon of 1.4 eV

33) Given below are two statements : one is labelled as Assertion (A) and other is labelled as Reason (R).
**Assertion (A)** : The electron in the hydrogen atom passes from energy level n = 4 to the n = 1 level. The maximum and minimum number of photon that can be emitted are six and one respectively.
**Reason (R)** : The photons are emitted, when electron make a transition from the higher energy state to the lower energy state.

(1) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of the **(A).**

(2) **(A)** is true but **(R)** is false.

(3) **(A)** is false but **(R)** is true.

(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of the **(A).**

34)

Energy is released in nuclear fission is due to
(a) Few mass is converted into energy
(b) Total binding energy (B.E.) of fragments is more than the B.E. of parental element
(c) Total B.E. of fragments is less than the B.E. of parental element
(d) Total B.E. of fragments is equals to the B.E. of parental element

(1) a,c

(2) a,b

(3) a,d

(4) all

35) When a photo sensitive surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated by light of wavelength 2λ, the stopping potential is $\frac{V}{3}$. The threshold wavelength for the surface is :

(1) $\frac{4\lambda}{3}$

(2) 4λ

(3) 6λ

(4) $\frac{8\lambda}{3}$

36) The de-Broglie wavelength of a neutron at 27°C is λ. What will be its wavelength at 927°C:-

(1) λ/2

(2) λ/3

(3) λ/4

(4) λ/9

37) Potential drop across a forward biased p-n junction diode is 0.7 V. If a battery of 4 V is applied, calculate the minimum resistance to be put in series, for the maximum current of 5 mA in the circuit.

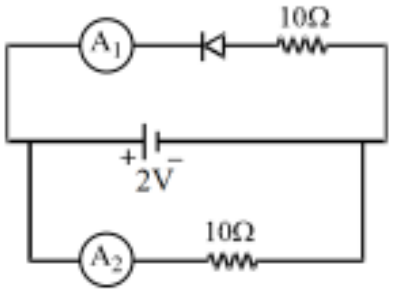
(1) 660 Ω

(2) 350 Ω

(3) 475 Ω

(4) 500 Ω

38) The reading of ammeters $A_1$ and $A_2$ in the given diagram will be (neglect the resistance of



ammeters)

(1) Zero, zero

(2) Zero, 0.1 A

(3) 0.1 A, zero

(4) Zero, 0.2 A

39) **Statement - 1 :** Doping of semiconductors increases its conductivity.
**Statement - 2 :** Forbidden energy gap between valence band and conduction band increases on increasing temperature.

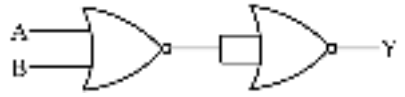
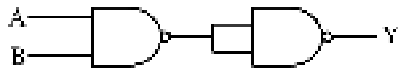
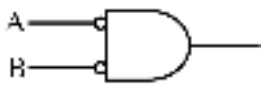
(1) Statement 1 is true, statement 2 is true.

(2) Statement 1 is false, statement 2 is true.

(3) Statement 1 is true, statement 2 is false.

(4) Statement 1 is false, statement 2 is false.

40) Match the columns :

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (A) |  | (i) | OR |
| (B) |  | (ii) | AND |
| (C) |  | (iii) | NOR |
| (D) |  | (iv) | NAND |

(1) A-(i), B-(ii), C-(iii), (D)-(iv)

(2) A-(i), B-(ii), C-(iv), (D)-(iii)

(3) A-(ii), B-(i), C-(iii), (D)-(iv)

(4) A-(iii), B-(iv), C-(i), (D)-(iii)

41) In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 volt battery and a high resistance of 11 kΩ are used. The figure of merit of the galvanometer is 60 μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ = 9 divisions when current flows in the circuit. The value of shunt resistance that can cause deflection of θ/2 is closest to :

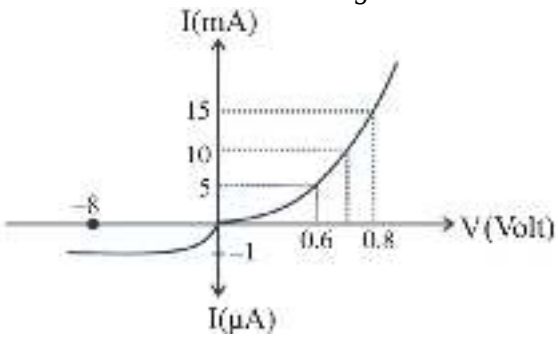
(1) 55 Ω

(2) 110 Ω

(3) 220 Ω

(4) 550 Ω

42) The I-V characteristics of a semiconductor diode is shown in the figure. The resistance of diode at V = –8 V and I = 10 mA are respectively




(1) 8M Ω , 20 Ω

(2) 8M Ω , 0 Ω

(3) 8K Ω , 20 Ω

(4) 20K Ω , 8 Ω

43) A travelling microscope is focussed on a mark on a piece of paper and the scale reading is 5.25 cm. A rectangular block of glass is placed on paper, and the microscope is raised and focused again on the mark, then microscope reading is 11.25 cm. Now a layer of lycopodium powder is sprinkled on the block of glass and viewed the top surface. The reading obtained is 23.25 cm. The refractive index of material of block is :

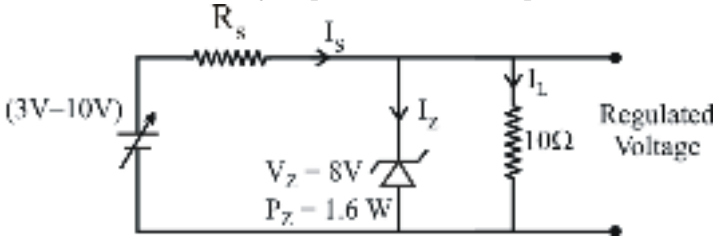
(1) 1.32

(2) 1.45

(3) 1.5

(4) 1.64

44) A zener diode of breakdown voltage 8 V is to be used as a voltage regulator across a fluctuating voltage, varying between 3V and 10 V. If the power rating of zener diode is 1.6 W, then the value of series resistance $R_s$ required for safe operation of diode will be :



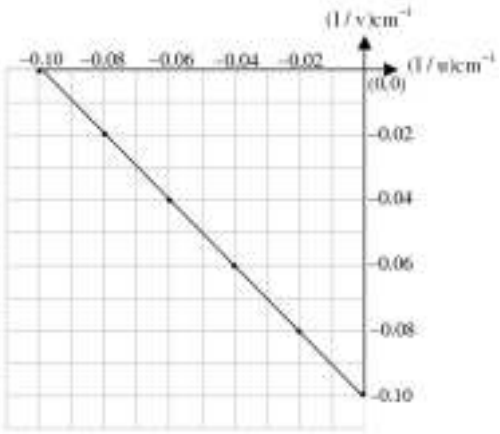

(1) 2 Ω

(2) 4 Ω

(3) 6 Ω

(4) 8 Ω

45) For a concave mirror, $\frac{1}{u}$ versus $\frac{1}{v}$ graph obtained by parallax method is as shown in figure. The graph is a straight line with slope –1. The estimated value of focal length of mirror is :




(1) 5cm

(2) 20cm

(3) –5cm

(4) –10cm

## CHEMISTRY

1) Which of the following has longest C–O bond length? (Free C–O bond length in CO is 1.128Å).

(1) $[Ni(CO)_4]$

(2) $[Co(CO)_4]^{\ominus}$

(3) $[Fe(CO)_4]^{2-}$

(4) $[Mn(CO)_6]^{+}$

2) **Statement-I :** Boron has highest value of boiling point and melting point in their respective group.
**Statement-II :** Boron is unreactive in crystalline form.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

3) The correct IUPAC name of $[Pt(NH_3)_4Cl_2]$ $[PtCl_4]$ is :-

(1) Tetraammine dichloro platinum (IV)
tetrachloro platinate (II)

(2) Dichlorotetra ammineplatinum (IV)
tetrachloro platinate (II)

(3) Dichlorotetraammineplatinum (II)
tetrachloro platinate(IV)

(4) Tetrachloro platinum (II)
dichloro tetraamine platinate (IV)

4) Write the correct formula for the
Dichloridobis(ethane –1,2-diamine) cobalt (III) ion

(1) $[CoCl_2(en)_2]$

(2) $[CoCl_2(en)_2]^+$

(3) $[CoCl_2(en)_2]^-$

(4) $[CoCl_2(en)]^+$

5) The complex showing a spin magnetic moment of 2.82 BM is-

(1) $[Ni(CO)_4]$

(2) $[NiCl_4]^{2-}$

(3) $[Ni(PPh_3)_4]$

(4) $[Ni(CN)_4]^{2-}$

6) In which of the following molecule orbital of different principle quantum number are used in the hybridization :-

(1) $[Ni(NH_3)_6]^{+2}$

(2) $[Ni(CO)_4]$

(3) $[PtCl_4]^{-2}$

(4) $K_4[Fe(F_6)]$

7)

Match column-I with column-II correctly.

| **Column-I (Elements/ion)** | | **Column-II (Color imparted to the flame)** | |
|---|---|---|---|
| (a) | K | (p) | Apple green |
| (b) | Na | (q) | Crimson red |
| (c) | Sr | (r) | Golden yellow |
| (d) | Ba | (s) | Violet |

| | **(a)** | **(b)** | **(c)** | **(d)** |
|---|---|---|---|---|
| (1) | r | s | q | p |
| (2) | s | p | q | r |
| (3) | s | r | q | p |
| (4) | p | r | q | s |

(1) 1

(2) 2

(3) 3

(4) 4

8) The wavelength of light absorbed is highest in

(1) $[Fe(CN)_6]^{3-}$

(2) $[Fe(NH_3)_6]^{3+}$

(3) $[Fe(NO_2)_6]^{3-}$

(4) $[FeCl_6]^{3-}$

9) $d^6$-octahedral low spin complex has following splitting energy :
($\Delta_0$ = CFSE for octahedral, P = pairing energy)

(1) $-\frac{2}{5}\Delta^0 + 2P$

(2) $-\frac{2}{5}\Delta^0 + P$

(3) $-\frac{12}{5}\Delta^0 + P$

(4) $-\frac{12}{5}\Delta^0 + 3P$

10) Which of the following complex will have two stereoisomeric forms ?
(I) $[Cr(NO_3)_3(NH_3)_3]$ (II) $K_3[Fe(C_2O_4)_3]$
(III) $[CoCl_2(en)_2]^+$ (IV) $[CoBrCl(Ox)_2]^{3-}$

(1) I only

(2) I and II

(3) III and IV

(4) All of these

11) Match **List-I** with **List-II** :

| | **List-I (Complexes)** | | **List-II (Types)** |
|---|---|---|---|
| (a) | $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$ | (i) | ionisation isomerism |
| (b) | $[Cr(NH_3)_6]$ $[Co(CN)_6]$ and $[Co(NH_3)_6]$ $[Cr(CN)_6]$ | (ii) | coordination isomerism |
| (c) | $[Co(NH_3)_5(SO_4)]Br$ and $[Co(NH_3)_5Br]SO_4$ | (iii) | linkage isomerism |

| | | | |
|---|---|---|---|
| (d) | $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$ | (iv) | solvate isomerism |

Choose the **correct answer** from the options given below :

(1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

12) **Assertion** : Tetrahedral complex do not exhibit geometrical isomerism. **Reason** : In tetrahedral complex all the four positions are identical.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion

(2) Both assertion and reason are correct and reason is not the correct explanation of assertion

(3) Assertion is correct and reason is incorrect.

(4) Assertion is incorrect and reason is correct.

13) $[Fe(en)_2(H_2O)_2]^{2+}$ + en → complex (x). The correct statement about the complex (x) it is:-

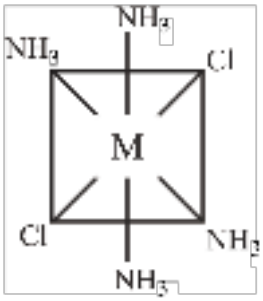
(1) a low spin complex

(2) diamagnetic

(3) shows geometrical isomerism

(4) (1) and (2) both

14) The isomer can be marked as :-




(1) Cis isomer

(2) Trans isomer

(3) d-isomer

(4) □-isomer

15) Hybridisation and shape of central metal atom of Wilkinson's catalyst is :-

(1) $dsp^2$, Square planar

(2) $sp^3$, Tetrahedral

(3) $sp^3d$, Trigonal bipyramidal

(4) $sp^3d^2$, Octahedral

16) Which statement is incorrect ?

(1) $Cr^{2+}$ is strongly reducing while $Mn^{3+}$ is strongly oxidising while both have $d^4$ configuration

(2) In 3d series only Cu has unique behaviour of having a positive $E^{\ominus}(M^{2+}/M)$

(3) $V_2O_5$ is amphoteric while $CrO_3$ and $Mn_2O_7$ is acidic

(4) Highest fluoride of Mn is $MnF_7$ while highest oxide is $Mn_2O_7$.

17) Which of the following is correct :-

(1) $CrO_4^{-2} \underset{pH=8}{\overset{pH=4}{\rightleftharpoons}} Cr_2O_7^{-2}$

(2) +6 oxidation state is shown by Ce, Tb

(3) $KMnO_4$ is violet due to d-d transition

(4) AgCl, $CrO_2Cl_2$ are coloured due to polaristion

18)

Which of the following is the incorrect match according to given property ?

(1) $NH_3 > PH_3 > AsH_3 > SbH_3$ (Bond angle)

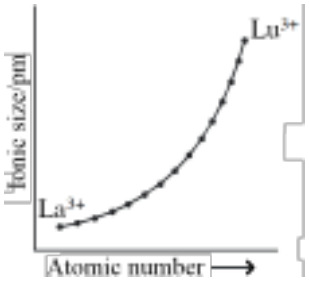
(2) $BiH_3 > SbH_3 > AsH_3 > PH_3 > NH_3$ (Boiling point)

(3) $NH_3 < PH_3 < AsH_3 < SbH_3$ (Acidic strength)

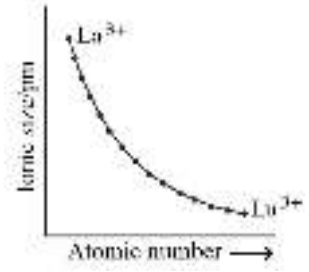
(4) $NH_3 > SbH_3 > AsH_3 > PH_3$ (Melting point)

19) Which of the following graphs shows correct ionic size order of +3 ions of lanthanides ?

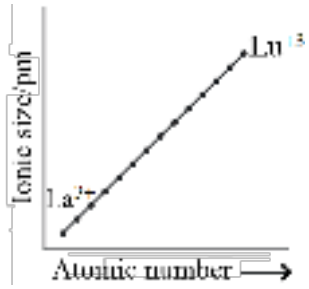
(1)

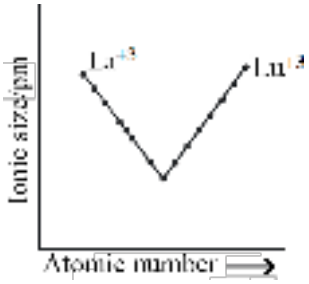



(2)




(3)



(4) 


20) Colour of Nickel chloride in aqueous solution is

(1) Pink
(2) Black
(3) Colourless
(4) Green

21) **Statement-I :** Increasing order of stability of oxides formed by halogen : I > Cl > Br.
**Statement-II :** $Br_2O$, $BrO_2$, $BrO_3$ are the least stable halogen oxides and exist only at low temperature.

(1) Both Statement I and Statement II are incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Statement I is correct but Statement II is incorrect.

22) Match the List-I with List-II

| | **List-I (Cation)** | | **List-II (Group reagent)** |
|---|---|---|---|
| A. | $Pb^{+2}$, $Cu^{+2}$ | i. | $H_2S$ + HCl |
| B. | $Al^{+3}$, $Fe^{+3}$ | ii. | $(NH_4)_2CO_3$ + $NH_4Cl$ |
| C. | $Co^{+2}$, $Ni^{+2}$ | iii. | $NH_4OH$ + $NH_4Cl$ |
| D. | $Ba^{+2}$, $Ca^{+2}$ | iv. | $H_2S$ + $NH_4OH$ |

Correct match is.
(1) A–iv; B–ii; C–iii; D–i
(2) A–i; B–iii; C–iv; D–ii
(3) A–iii; B–i; C–iv; D–ii
(4) A–i; B–iii; C–ii; D–iv

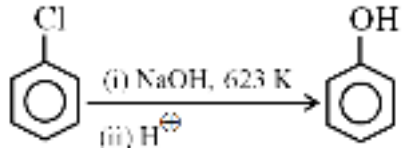
23)

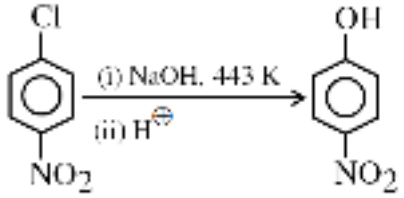
Potassium thiocyanate solution react with ferric chloride to give :

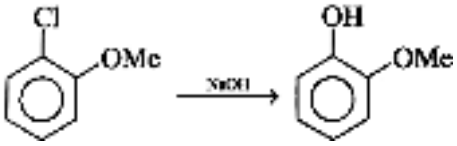
(1) Pink colour
(2) Blood red colour
(3) Green colour

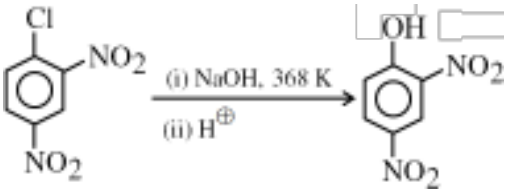
(4) Deep blue colour

24) Which reaction is not feasible?

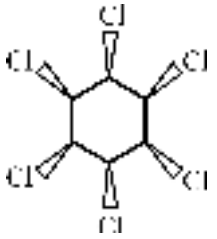
(1) 

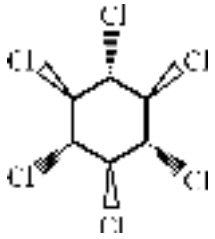

(2) 

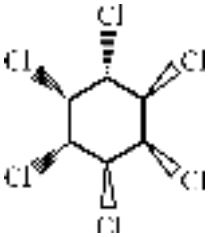

(3) 


(4) 


25) Arrange the following in decreasing order of $E_2$ reaction (dehydrohalogenation):

(X) 

(Y) 

(Z) 

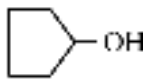
(1) X > Y > Z

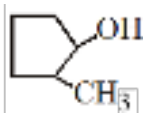
(2) X > Z > Y

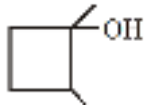
(3) Z > X > Y

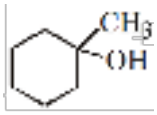
(4) Y > X > Z

26) An organic compound A (molecular formula $C_6H_{12}O$) on treatment with $H_2SO_4$ produce Alkene, which on oxidative ozonolysis give a molecule ($C_6H_{10}O_2$), which give positive Iodoform test, Find out structure of "A".

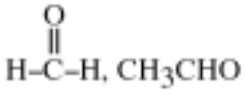
(1) 

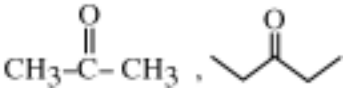
(2) 

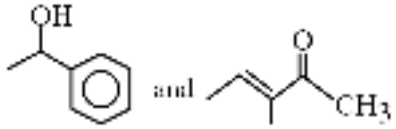
(3) 

(4) 

27) Which of the following pair is not differentiated by iodoform test?

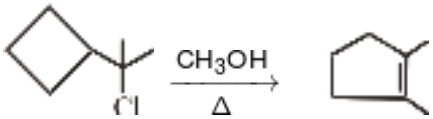
(1) 

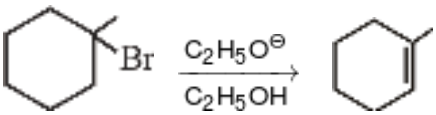
(2) 

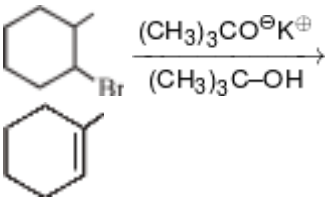
(3) 

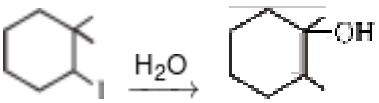
(4) $CH_3$–$CH_2$–OH, $CH_3$–$CH_2$–$CH_2$–OH

28) Which of the following reaction does not represent major product ?

(1) 

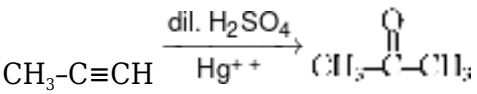
(2) 

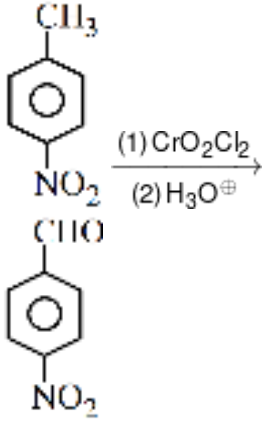
(3) 

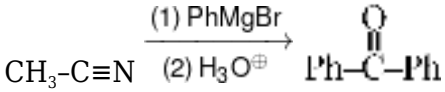
(4) 

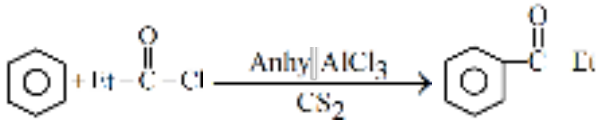
29)

Incorrect reaction is :-

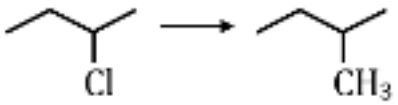
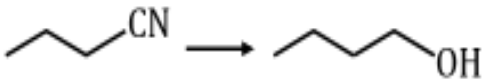
(1) 

(2) 

(3) 

(4) 

30) Match the column-

| | Column-I (Reaction) | | Column-II (Reagents) |
|---|---|---|---|
| (a) |  | (p) | $H_3O^+$ ; $LiAlH_4$ |
| (b) |  | (q) | $CH_3MgX$ |
| (c) |  | (r) | $NaBH_4$ ; $PCl_3$ |
| (d) |  | (s) | Red P + HI |

(1) a → s ; b → q ; c → r ; d → p

(2) a → q ; b → s ; c → r ; d → p

(3) a → s ; b → r ; c → q ; d → p

(4) a → s ; b → q ; c → p ; d → r

31) Which statement is incorrect for Hoffmann Bromamide reaction ?

(1) Number of carbon atom in product amine is one less than amide used as reactant

(2) R-CONH-Br & R-N=C=O are formed as intermediate

(3) 4 moles of KOH consumed in reaction

(4) Aniline cannot be prepared by this method

32) When the Lassaigne extract is treated with ferric chloride a blood red colour appears. Which

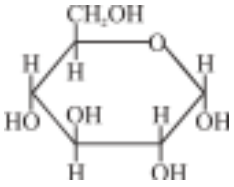
elements present in organic compound-

(1) Phosphorous
(2) Nitrogen
(3) Sulphur
(4) Both Nitrogen & sulphur

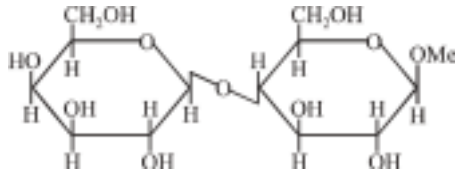
33) **Assertion :** Glycerol is separated from spent lye by vacuum distillation.
**Reason :** Glycerol is decomposed at their common boiling point.

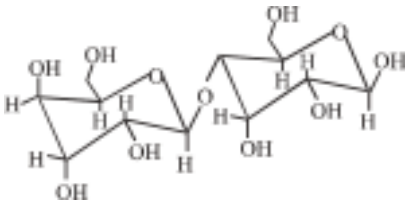
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

34) Which of the following is a non-reducing carbohydrate:-

(a) 

(b) 

(c) 

(d) 

(1) Only a, b, d
(2) Only b, c, d
(3) Only b, c
(4) Only c, d

35) Match List-I with List-II.

| | **List-I (Vitamins)** | | **List-II (Deficiency diseases)** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| (a) | Vitamin-$B_1$ | (i) | Scurvy |
| (b) | Vitamin-$B_2$ | (ii) | Beri-Beri |
| (c) | Vitamin-C | (iii) | Pernicious anaemia |
| (d) | Vitamin-$B_{12}$ | (iv) | Cheilosis |

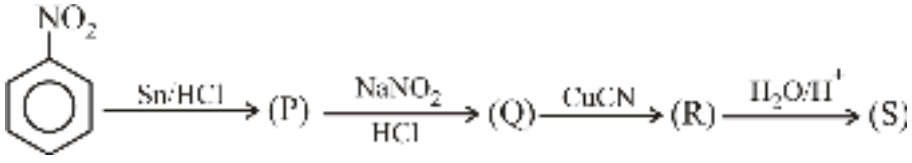
Choose the correct option from the options given below :

(1) a-iv, b-ii, c-i, d-iii

(2) a-ii, b-iv, c-i, d-iii

(3) a-iv, b-ii, c-iii, d-i

(4) a-ii, b-iii, c-i, d-iv

36) 

For the above sequence following statement is false :

(1) (R) is benzonitrile

(2) (P) to (Q) is diazotisation reaction

(3) (P) is a primary amine

(4) (S) on sodalime decarboxylation produce cyclohexane

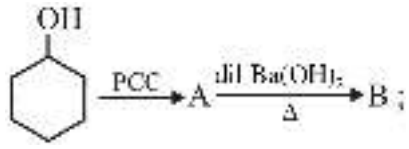
37) Select the correct option.

(1) Boiling point of alcohols and phenols increases on increasing number of carbon

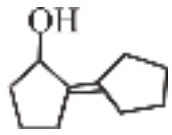
(2) Chloroform in presence of light & oxygen does not form phosgene gas

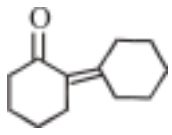
(3) Boiling point of ether is greater than alcohol when molecular weight is almost same

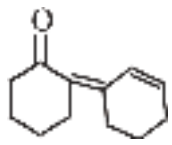
(4) Haloalkanes are completely soluble in water

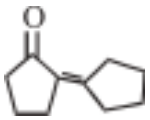
38) 

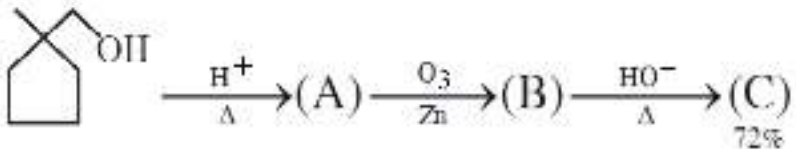
final product 'B' will be :

(1) 

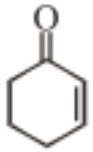
(2) 

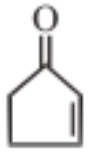
(3) 

(4) 

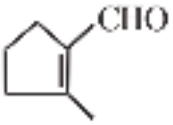
39) 


Product (C) is :-

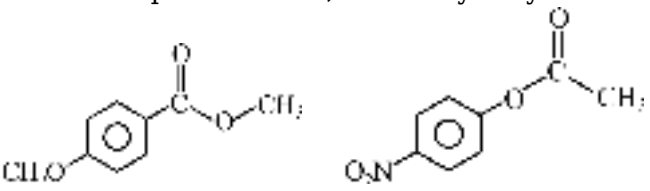
(1) 

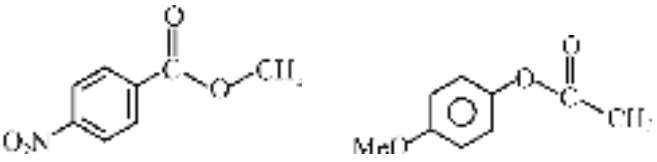
(2) 

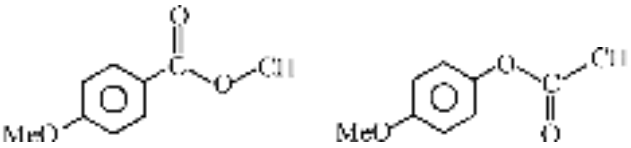
(3) 

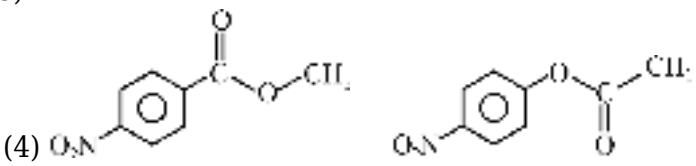

(4) 


40) Select the pair in which, rate of hydrolysis is fastest :-

(1) 

(2) 

(3) 

(4) 

(1) 1

(2) 2

(3) 3

(4) 4

41) Mixture of PhCHO and HCHO is treated with NaOH, then cannizzaro reaction involve :-
(A) Reduction of HCHO
(B) Oxidation of HCHO
(C) Reduction of PhCHO
(D) Oxidation of Ph–CHO

(1) A,C

(2) A,D

(3) B,C

(4) B,D

42) The product obtained by the alkaline hydrolysis of $C_2H_5$—NCO when treated with t–butyl magnesiumbromide, the compound obtained will be

(1) t–butylamine

(2) n–butylamine

(3) Isobutane

(4) n–butane

43) Which of the following compounds after reacting with benzene suphonyl chloride is soluble in alkali solution :-

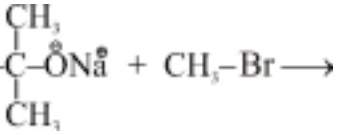
(1) $CH_3$–$CH_2$–$CH_2$–$NH_2$

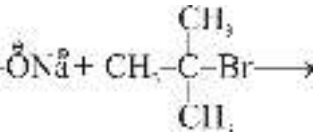
(2) $CH_3$–$CH_2$–NH–$CH_3$

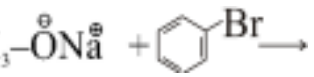
(3) $CH_3CH_2$–OH

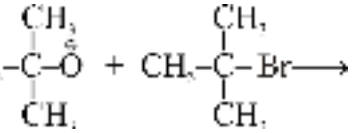
(4) $(CH_3)_3N$

44) Which of the following reaction will produce ether as major product?

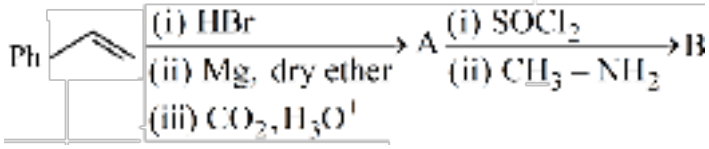
(1) 

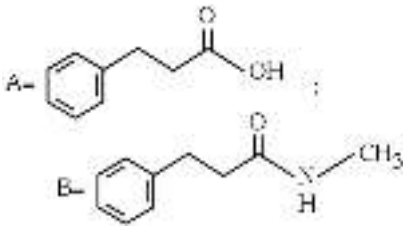
(2) 

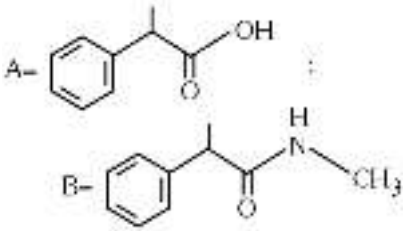
(3) 

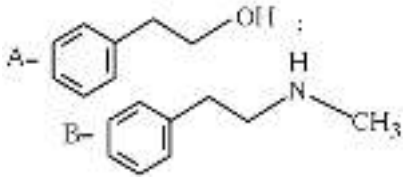
(4) 

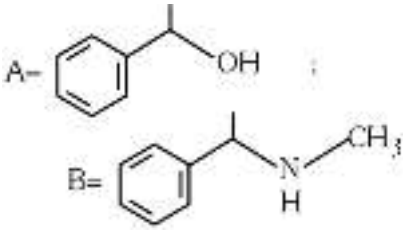
45) The products A and B in the following reaction sequence are :




(1)



(2)



(3)



(4)



## BIOLOGY

1) Read column-I, II and III and identify their correct match :

| | **Column-I** | | **Column-II** | | **Column-III** |
|---|---|---|---|---|---|
| (A) | Dihybrid cross | (P) | $I^A$, $I^B$ & i | (i) | 28s, 18s and 5.8s |
| (B) | Multiple allele | (Q) | Law of independent assortment | (ii) | Exon & Intron both |
| (C) | RNA polymerase-I | (R) | r-RNA | (iii) | 9 : 3 : 3 : 1 |
| (D) | Primary transcripts | (S) | hn RNA | (iv) | ABO blood groups |

(1) (A)-(Q)-(iii), (B)-(P)-(iv), (C)-(R)-(i), (D)-(S)-(ii)

(2) (A)-(P)-(v), (B)-(Q)-(iii), (C)-(S)-(iv), (D)-(R)-(iii)

(3) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(4) (A)-(S)-(iv), (B)-(R)-(iii), (C)-(Q)-(ii), (D)-(P)-(i)

2) Consider the following statements and choose the correct options:
(A) In incomplete dominance, the $F_1$ had a phenotype that did not resemble either of the two parents and was in between the two.
(B) In co-dominance, their is blending of characters.
(C) Dominance is an autonomous feature of a gene or the product that it has information.
(D) In an individual (2n) only two alleles of –a gene can be present, multiple alleles can be found only when population studies are made.
(E) Gene which controls more than one character is called multiple alleles.

(1) A, B and E

(2) B, C and D

(3) A and D

(4) B and D

3)

Some events of protein synthesis in prokaryotes are given below –
(A) Binding of first charged t-RNA to ribosomal P site.
(B) Activation of amino acids.
(C) Binding of charged t-RNA to ribosomal A site during elongation
(D) Binding of smaller subunit of ribosome to m-RNA.
(E) Activity of releasing factors.
In which of the following option these events are arrange in specific sequence :-

(1) B → A → E → D → C

(2) A → B → E → C → D

(3) B → E → C → A → D

(4) B → D → A → C → E

4) Consider the following four statements (a-d) and select the option which includes all the correct ones only :-
(a) The codon is read in mRNA in a contiguous fashion.
(b) UAG codon has dual function in protein synthesis.
(c) In actual structure, the tRNA is a compact molecule which looks like inverted 'L'.
(d) 28's' rRNA in bacteria behave as ribozyme.

(1) b, c and d

(2) a, b and d

(3) a, c and d

(4) a and c

5) Correctly match column-I with column-II :-

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (A) | $1^{st}$ Chromosome | (i) | 1.4 million locations |
| (B) | SNPs | (ii) | 2968 genes |
| (C) | Dystrophin gene | (iii) | 231 genes |

| | | | |
|---|---|---|---|
| (D) | Y-Chromosome | (iv) | 2.4 million bp |

**Options :-**

(1) A-i, B-ii, C-iv, D-iii

(2) A-ii, B-iv, C-i, D-iii

(3) A-ii, B-i, C-iv, D-iii

(4) A-ii, B-i, C-iii, D-iv

6) Given below are two statements :
**Statement I :** DNA polymerases catalyse polymerisation only in one direction, that is 5' → 3'
**Statement II :** During replication of DNA, on one strand the replication is continuous while on other strand it is discontinuous.
In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

7) Find out incorrect statement with respect to DNA structure.

(1) Two polynucleotide chains in a DNA are linked through hydrogen bonds.

(2) Two adjacent nucleotides in each polynucleotide chain of DNA, are linked through 3' - 5' phosphodiester linkage.

(3) Guanine is bonded with cytosine with three hydrogen bonds while adenine forms two hydrogen bonds with 5-methyl uracil.

(4) The pitch of DNA helix is 34 nm and there are roughly 10 bp in each turn, hence the distance between two adjacent bps in a helix is approximately 3.4 nm.

8) Match column-I to column-II :-

| | **Column-I** (Codon) | | **Column-II** (Amino acid) |
|---|---|---|---|
| (i) | AUG | (a) | Valine |
| (ii) | GUG | (b) | Methionine |
| (iii) | AAA | (c) | Phenylalanine |
| (iv) | UUU | (d) | Lysine |

**Options :-**

(1) i-a, ii-b, iii-c, iv-d

(2) i-a, ii-d, iii-c, iv-b

(3) i-b, ii-a, iii-d, iv-c

(4) i-c, ii-a, iii-d, iv-b

9) Which of the following feature provide stability to helical structure of DNA ?

(1) Presence of hydrogen bonds between two strands.
(2) The plane of one base pair stacks over the other in double helix.
(3) Parallel polarity of two strands.
(4) Both (1) and (2)

10) In lac-operon, repressor protein binds to :-

(1) Regulator gene
(2) Operator gene
(3) Promoter gene
(4) Structural gene

11) Which of the following pair is correctly matched ?

(1) Ribosomal RNA – Carries amino acid to the site of protein synthesis.
(2) Transcription – Process by which protein is synthesised.
(3) Translation – Process by which DNA carries information from the nucleus to ribosomes.
(4) Anticodon – Site of a tRNA molecule that binds to the mRNA molecule.

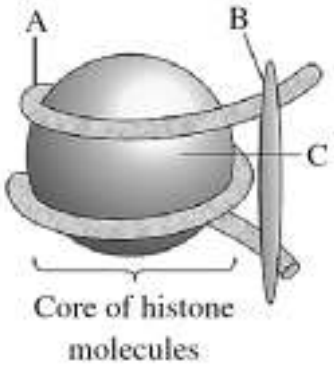
12) During DNA replication one strand is formed continuous and second strand is formed discontinuous because :-

(1) DNA synthesis is bidirectional
(2) DNA replication is semiconservative
(3) DNA chain formation take place in 5' to 3' direction only
(4) DNA replication is discontinuous

13) Repetitive DNA are separated from bulk genomic DNA as different peak during density gradient centrifugation. The bulk DNA forms a major peaks and the other small peaks are referred to as :-

(1) Repetitive DNA
(2) SNPs (Snips)
(3) Taq-DNA
(4) Satellite DNA

14) In given diagram identify B :




(1) $H_1$ - Histone

(2) DNA

(3) Histone octamer

(4) Nucleosome

15) Given below are two statements: one is Assertion (A) and other is Reason (R):
**Assertion (A):** In Incomplete dominance phenotype of $F_1$ resembled either of the two parents.
**Reason (R):** In Incomplete dominance dominant allele was not completely dominant over recessive allele.
In the light of the above statements choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

16) Select the correct statements about Thalassemia :
(A) It is an autosome linked recessive blood disease.
(B) α-Thalasemmia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent.
(C) β-Thalasemmia is controlled by single gene HBB on chromosome 11 of each parent.
(D) Thalassemia is a qualitative problem of synthesising an incorrectly functioning globin.

(1) A and B only

(2) C and D only

(3) A, B and C only

(4) B, C and D only

17) Match List-I with List-II

| | **List-I (Type of cross)** | | **List-II (Phenotypic ratio)** |
|---|---|---|---|
| (A) | Monohybrid test cross | (I) | 9 : 3 : 3 : 1 |
| (B) | Dihybrid test cross | (II) | 1 : 2 : 1 |
| (C) | Incomplete dominance | (III) | 1 : 1 : 1 : 1 |
| (D) | Dihybrid cross | (IV) | 1 : 1 |

Choose the correct answer from the options given below :

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-II, D-I

(4) A-IV, B-III, C-I, D-II

18) A normal female whose father was colourblind marries a man whose mother was also colourblind, what are the chance of her progeny having colourblindness?

(1) 25%

(2) 100%

(3) 75%

(4) 50%

19) Select incorrect option :-

(1) Allele - Alternative forms of a gene.

(2) Law of dominance - The alleles show blending.

(3) Genotype - Genetic constitution of an organism.

(4) Law of purity of gametes - each gamete receives only one factor of a pair.

20) In $F_2$ generation a ratio 1 : 4 : 6 : 4 : 1 is obtained instead of 9 : 3 : 3 : 1 when two pair of gene are considered, it indicates.

(1) Pleiotropic gene

(2) Quantitative inheritance

(3) Incomplete dominance

(4) Co-dominance

21) In Morgan's experiment with *Drosophila* the genes 'w' and 'y' were tighly linked and percent recombination showed by them was :-

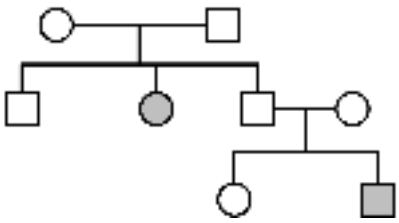
(1) 37.2

(2) 1.3

(3) 62.8

(4) 98.7

22) Following pedigree represents what type of genetic disorder :-



(1) Autosomal dominant

(2) X-linked recessive

(3) Autosomal recessive

(4) X-linked dominant

23) Turner's syndrome is denoted by -

(1) 44 + XXX

(2) 44 + YY

(3) 44 + XO

(4) 47 + XXY

24) Which one of the following is **not** appropriately defined ?

(1) Amensalism is a relationship in which one species is harmed whereas the other is unaffected.
(2) Predator is an organism that catches, kills & eats other organisms.
(3) Parasite is an organism which may live inside or on the body of the other organism.
(4) Female *Anopheles* mosquito is considered as parasite.

25) Which adaptation of parasites help to sustain their life in host body ?

(1) Abundance of sense organs
(2) Presence of adhesive organs or suckers to cling on the host
(3) Well developed digestive system
(4) Low reproductive capacity

26) **Assertion :** If a predator overexploits its prey, then the prey and predator both may become extinct.
**Reason :** Predators in nature are 'prudent'.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

27)

Match the following columns correctly :

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Epiphyte on its host | (i) | Mutualism |
| (B) | Plant-Seed dispersers | (ii) | Predation |
| (C) | Plant-Herbivore | (iii) | Commensalism |
| (D) | Human and Liver Fluke | (iv) | Parasitism |

(1) A-i, B-ii, C-iii, D-iv
(2) A-ii, B-i, C-iv, D-iii
(3) A-iii, B-i, C-ii, D-iv
(4) A-iii, B-i, C-iv, D-ii

28) How many examples in the list given below are related with the interaction, in which either one or both the interacting species get harmed?
Dog-Ticks, Cattle-Egret bird, Sea anemone-Clown fish, Human Liver Fluke-Snail, Lichen, Mycorrhiza, Copepod-Fish, Fig-Wasp

(1) Three
(2) Four
(3) Five

(4) Six

29) Which of the following pairs of characters decrease the population density?

(1) Mortality and Emigration
(2) Natality and Immigration
(3) Immigration and Mortality
(4) Emigration and Natality

30) If after exponential increase, population growth rate declines and then growth become stagnant. The growth curve is :

(1) S-shaped
(2) J-shaped
(3) Straight line
(4) Circular

31) Which one of the following is **incorrect**?

(1) Population ecology links ecology to population genetics and evolution
(2) Tiger census in our national park is often based on pug marks & fecal Pellets
(3) Biomass is generally the most appropriate measure of population density
(4) Sex ratio is an attribute of population

32) If b = 0.5, N = 10000, d = 0.4, then find $\frac{dN}{dt}$ for a population.

(1) 2000
(2) 1000
(3) 200
(4) 100

33) Usually which one of the following has the largest number in a grazing food chain?

(1) Producers
(2) Primary consumers
(3) Secondary consumers
(4) Decomposers

34) Which of the following is considered as structural & functional unit of nature ?

(1) Organism
(2) Ecosystem
(3) Community
(4) Population

35) Which of the following ecosystems shows inverted pyramid of number?

(1) Grassland ecosystem
(2) A tree ecosystem
(3) Pond ecosystem
(4) Aquatic ecosystem

36) Which of the following is **not** correct regarding ecological pyramids ?

(1) Saprophytes are not given any place
(2) They assume a simple food chain
(3) It does not take into account the same species belonging to two or more trophic levels
(4) They accommodate food web

37) Match the **column–I** with **column–II** :-

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (A) | Accumulation of dark coloured amorphous substance | (i) | Fragmentation |
| (B) | Water soluble nutrients go down into the soil horizon | (ii) | Humification |
| (C) | Release of inorganic nutrients by microbial activities | (iii) | Mineralisation |
| (D) | Break down of detritus into smaller particles | (iv) | Leaching |

| | **A** | **B** | **C** | **D** |
|---|---|---|---|---|
| (1) | (ii) | (iii) | (iv) | (i) |
| (2) | (ii) | (iv) | (iii) | (i) |
| (3) | (i) | (iii) | (iv) | (ii) |
| (4) | (iv) | (i) | (ii) | (iii) |

(1) 1
(2) 2
(3) 3
(4) 4

38) **Statement-I :** Each trophic level has a certain mass of living material at a particular time called as the standing crop.
**Statement-II :** The number of trophic level is not restricted in grazing food chain. [GFC]

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Statement-I & Statement-II both are incorrect.

39) **Assertion (A) :** Rate of biomass production by plants or the primary productivity varies in different type of ecosystems.
**Reason (R) :** Rate of biomass production or primary productivity depends on the plant species inhabiting a particular area, variety of environmental factors, availability of nutrients and photosynthetic capacity of plants.
Select the correct option with respect to above statements.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

40) Most diverse among vertebrates are :-

(1) Insects

(2) Fishes

(3) Birds

(4) Reptiles

41)

To describe the combined diversity at all the levels of biological organisation, 'biodiversity' term is popularised by-

(1) David Tilman

(2) Paul Ehrlich

(3) Robert May

(4) Edward Wilson

42) Which one of the following species in the world facing the minimum threat of extinction?

(1) Birds

(2) Mammals

(3) Amphibians

(4) Gymnosperm

43) There are four major causes ( the evil quartet) of biodiversity losses, with respect to them, which of the following statement is **correct** ?

(1) The degradation of habitat by pollution may also threatens the survival of many species.

(2) Over exploitation caused the extinction of carrot grass from India

(3) Alien species invasions decrease the threat to native species.

(4) When a species becomes extinct, the plant and animal species associated with it in an facultative way also become extinct.

44) **Assertion :** If we want to conserve the whole ecosystem and its biodiversity at all levels, the desirable approach is in-situ conservation.
**Reason :** In-situ conservation approach includes some of the most scientifically advanced methods like tissue culture etc.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

45) ____**(A)**___ are found in Khasi and Jaintia, Hills of Meghalaya, Aravali hills of ____**(B)**___, western ghat regions of Karnataka and Maharashtra and the Sarguja, Chandra and Baster areas of ____**(C)**___. In Meghalaya, the sacred groves are the last refuges for a large number of rare and threatened species. India has ____**(D)**___ Biosphere reserves and 90 national parks.
What does **A,B,C,D** represent respectively in above paragraph?

(1) Sacred groves, Madhya Pradesh, Rajasthan, 448

(2) Sacred groves, Rajasthan, Madhya Pradesh, 14

(3) Biosphere reserve, Rajasthan, Uttar Pradesh, 14

(4) Zoological park, Madhya Pradesh, Rajasthan, 34

46) An antibiotic resistance gene of plasmid vector which don't get inactivated due to insertion of alien DNA, helps in the selection of :-

(1) Transformants

(2) Recombinants

(3) Non-recombinants

(4) (2) & (3) both

47) Which of the following DNA sequences can be a recognition site of a restriction endonuclease enzyme?

(1)
A A G G
T T C C

(2)
A A T T C C G G
T T A A G G C C

(3)
C T C G A G
G A G C T C

(4)
G G T T C G
C C A A G C

48) $p^{BR322}$ vector is cut by **Pst I** enzyme and desired gene is ligated in vector at the recognition site of **Pst I** and recombinant DNA is formed, which is transferred into the host cell.
Select the correct statement:

(1) Recombinant cells will grow in ampicillin containing medium.

(2) Non-transformant cells will grow in tetracycline containing medium.

(3) Recombinant cells will grow in tetracycline containing medium.

(4) Non-transformant cells will grow in ampicillin containing medium.

49) Match Column-I and Column-II and select the right option

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| I. | r-DNA technology | A. | Vector |
| II. | Cloning vehicles | B. | Restriction enzyme |
| III. | Molecular scissors | C. | Electrophoresis |
| IV. | Agarose gel | D. | Genetic engineering |

(1) I-D, II-A, III-B, IV-C

(2) I-A, II-D, III-B, IV-C

(3) I-D, II-A, III-C, IV-B

(4) I-B, II-A, III-D, IV-C

50) Arrange the steps of rDNA technology in correct order:-
I. Extraction of the desired gene product.
II. Amplification of gene of interest by PCR.
III. Isolation of desired DNA fragment.
IV. Ligation of DNA fragment into vector.
V. Insertion of rDNA into host.

(1) I, II, III, IV, V

(2) V, IV, III, II, I

(3) III, II, IV, V, I

(4) III, IV, II, I, V

51) In agarose gel electrophoresis :-

(1) RNA migrates towards negative electrode

(2) Larger molecules migrate faster than smaller molecules

(3) Smaller molecules migrate faster than larger molecules

(4) DNA do not show migration

52) Consider the following statement :-
(A) Presence of more than one recognition sites of a restriction endonuclease within the vector will complicate the gene cloning.
(B) ori site of vector is responsible for controlling the copy number of the linked DNA.
Select the correct option:-

(1) Both (A) and (B) are true

(2) (A) is true but (B) is false

(3) Both (A) and (B) are false

(4) (A) is false and (B) is true

53) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
**Assertion (A) :** Insulin cannot be effectively administered orally to diabetic patients.
**Reason (R) :** Insulin is a protein that gets broken down by digestive enzymes in the stomach and intestines.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true.

54) Which of the following statement are incorrect ?
(a) RNAi takes place in all eukaryotic organisms as a method of cellular defence.
(b) C peptide along with A and B peptides are present in mature insulin.
(c) In 1990, the first transgenic cow, Rosie, produced human protein-enriched milk.
(d) Recombinant DNA technology serve the purpose of early diagnosis.
(e) 27 Documented Basmati rice varieties are grown in India.

(1) b, c, d

(2) a, b, c

(3) a, d, e

(4) b and c only

55) **Assertion :** Biopiracy is the term used to refer to the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment.
**Reason :** 1997, an American company got patent rights on Basmati rice through the US Patent and Trademark Office

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

56) Which of the following statements are correct ?
A. Genetic modification has made crops intolerant to abiotic stress.
B. A nematode *Meloidegyne incognita* infects the leaves of tobacco plant.
C. Gene therapy is a collection of methods that allows correction of a gene defect.
D. Today transgenic models exist for many human disease such as cancer.
E. Toxicity testing in transgenic animals will allow us to obtain results in less time.
Choose the most appropriate answer from the options given below.

(1) B, C and D only

(2) C, D and E only

(3) A, B and E only

(4) A, B and D only

57) **Statement-A** : At present about 30 recombinant therapeutics have been approved for human use the world over.
**Statement-B** : 27 documented varieties of Basmati rice are grown in India.

(1) Both A and B are true

(2) Only A is true

(3) Only B is true

(4) Both A and B are false

58) Match column-I with Column-II and select the answer from the codes given below.

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Biopiracy | (i) | Effort to correct the non-functional gene |
| (B) | Biopatent | (ii) | Gene silencing |
| (C) | Gene therapy | (iii) | illegal exploitation of biological materials |
| (D) | RNAi | (iv) | Right granted for biological entities |

(1) A - (iv), B - (ii), C - (i), D - (ii)

(2) A - (ii), B - (iv), C - (i), D - (iii)

(3) A - (iii), B - (iv), C - (i), D - (ii)

(4) A - (iii), B - (iv), C - (ii), D - (i)

59) Which of the following is not a critical research area in biotechnology?

(1) Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.

(2) Creating optimal conditions through engineering for a catalyst to act.

(3) Downstream processing technologies to purify the protein/organic compound.

(4) Protein transfer technology using artifical vectors in strains of *E. coli.*

60) A single stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography. It will help studying diseases by getting following results :-

(1) The clone having the mutated gene will hence not appear on the photographic film.

(2) The clone having normal gene will hence not appear on the photographic film.

(3) The clone having the mutated gene will hence appear on the photographic film.

(4) The clone having the mutated gene doesn't appear on photographic film as the probe has complementarity with it.

61) Read the following sentences.
(a) In elution the separated bands of DNA are cut out from agarose gel and extracted from the gel piece
(b) *E. coli* cloning vector PBR-322 have several restriction sites, an ori, antibiotic resistance genes and rop.

(c) The downstream processing and quality control testing vary from product to product.
(d) Competent bacterial cells take up the plasmid.

(1) All are incorrect

(2) Except d, all are correct

(3) Only d is correct

(4) All are correct

62) Match the column :

| | **Column-I** | | **Column-II** | | **Column-III** |
|---|---|---|---|---|---|
| (a) | RNA interference | (i) | ADA deficiency | (t) | Kill insects |
| (b) | *Bacillus thuringiensis* | (ii) | Gene silencing mechanism | (u) | Hereditary disease |
| (c) | First clinical gene therapy | (iii) | Heat stable enzyme | (v) | DNA amplification |
| (d) | Taq polymerase | (iv) | insecticidal protein production | (w) | Cellular defence |

(1) a-ii-w, b-iii-t, c-iv-u, d-i-v

(2) a-iii-v, b-iv-w, c-ii-u, d-i-t

(3) a-ii-w, b-iv-t, c-i-u, d-iii-v

(4) a-ii-w, b-iv-t, c-iii-u, d-i-v

63) The ladybird and dragon flies are useful to get rid of ___a___ and ___b___ respectively.

(1) a-Aphids, b-Mosquitoes

(2) a-Mosquitoes, b-Butterfly

(3) a-Butterfly, b-*Bacillus*

(4) a-*Bacillus*, b-Mosquitoes

64) Read the following statements :-
(A) Cyclosporin A is used as a clot buster
(B) Whisky, brandy & rum are produced by distillation of fermented broth
(C) Baculoviruses are used as bio control agents
(D) The biogas production technology in India in developed by GEAC
Which of the above statements are correct ?

(1) A & B

(2) B & C

(3) C & D

(4) A & D

65)

Which of the following microbes is **NOT** involved in the preparation of household products?

A. *Aspergillus niger*
B. *Lactobacillus*
C. *Trichoderma polysporum*
D. *Saccharomyces cerevisiae*
E. *Propionibacterium sharmanii*

(1) A and B only
(2) A and C only
(3) C and D only
(4) C and E only

66) Toddy is made by fermenting sap from :

(1) Rice
(2) Palm
(3) Wheat
(4) Maize

67) **Statement - 1 :** Butyric acid is produces by *Clostridium butylicum*.
**Statement - 2 :** Lactic acid is produced by *Lactobacillus*.

(1) Both statement are correct.
(2) Both statement are incorrect.
(3) Statement 2 correct statement 1 incorrect.
(4) Statement 1 correct statement 2 incorrect.

68) Which of the following is an auto immune disease?

(1) Myasthenia gravis
(2) SCID
(3) Asthma
(4) Alzheimer disease

69) Identify the correct pair representing the causative agent of typhoid fever and the confirmatory test for typhoid.

(1) *Salmonella typhi* / Widal test
(2) *Plasmodium vivax* / UTI test
(3) *Streptococcus pneumoniae* / Widal test
(4) *Salmonella typhi* / Anthronc test

70) In malignant tumors, the cells proliferate, grow rapidly and move to other parts of the body to form new tumors. This stage of disease is called:

(1) Metagenesis
(2) Metastasis
(3) Teratogenesis

(4) Mitosis

71) Diseases are broadly grouped into infectious and non-infectious diseases. In the list given below, identify the infectious diseases.
(i) Cancer (ii) Influenza
(iii) Allergy (iv) Small pox

(1) i and ii
(2) ii and iii
(3) iii and iv
(4) ii and iv

72) Opportunistic infections appear in AIDS patient due to low level of -

(1) B-lymphocyte
(2) T-killer cell
(3) Natural killer
(4) T-helper

73) Read the following statement.
**Statement-I :** HIV is not an congenital disease.
**Statement-II :** HIV is a retrovirus have 2 identical molecule of ssDNA.
Choose the correct option -

(1) Both statement-I and statement-II wrong.
(2) Statement-I is right but statement-II is wrong.
(3) Statement-I is wrong but statement-II is right.
(4) Both statement-I and statement-II are right.

74)

In the given list how many drugs are obtained from flowering plants ?
Morphine, Marijuana, Charas, Datura, Ganja

(1) 5
(2) 4
(3) 3
(4) 2

75) Fill in the blanks :
After getting in to the body of the person, the virus enters into ____A____ where RNA genome of the virus replicates to form viral ____B___ with the help of the enzyme ___C___.

(1) A-Macrophage B-RNA C-Reverse transcriptase
(2) A-TH-cell B-DNA C-Restriction endonuclease
(3) A-Macrophase B-DNA C-Reverse transcriptase
(4) A-Tk-cell B-RNA C-Restriction endonuclease

76) Disease of human being caused by protozoan parasites are :
(a) Malaria
(b) Amoebiasis
(c) Ascariasis
(d) Filariasis

(1) A, B, C
(2) A, B
(3) A, B, D
(4) B, C, D

77) In pneumonia pathogen infects the alveoli of the lungs. As a result of this alveoli get filled with fluid leading to severe problem in respiration. The symptoms of pneumonia are fevers, chills, cough and headache.
Given below are two statements about penumonia:
**Statement-I:** Pneumonia is caused by haemophilus influenzae.
**Statement-II:** Pneumonia is a vector borne disease.

(1) Both statement-I & II are correct
(2) Both statement-I & II are incorrect
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct

78) Given table gives the classification of disease

| ***Disease*** | ***Pathogen*** |
|---|---|
| Diphtheria | i |
| Malaria | ii |
| iii | Fungal |
| iv | Viral |

(1) i-Bacterial, ii-Protozoans, iii-Ringworm, iv-AIDS
(2) i-Protozoan, ii-Bacterial, iii-Plague, iv-AIDS
(3) i-Viral, ii-Protozoan, iii-Ringworm, iv-Dengue
(4) i-Protozoan, ii-Viral, iii-Axariasis, iv-Dengue.

79) Match the columns related with drugs, their examples and features.

| **Column-I** | **Column-II** |
|---|---|
| ____(i)____ | It is a depressant and slow down body functions |
| Very effective sedative and pain killer | ____(ii)____ |

| | |
|---|---|
| ____(iii)____ | It interface with the transport of the neurotransmitter dopamine |
| ____(iv)____ | Drug obtained from fungi *Claviceps purpurca* |

(1) (i)-Heroin, (ii)-Morphine, (iii)-Coke, (iv)-LSD

(2) (i)-Cannabinoids, (ii)-Coke, (iii)-LSD, (iv)-Morphine

(3) (i)-Ganja, (ii)-Coke, (iii)-Morphine, (iv)-Heroin

(4) (i)-LSD, (ii)-Morphine, (iii)-Heroin, (iv)-Bhang

80) Which option is not an example of analogous organs ?

(1) Wings of butterfly and birds

(2) Eyes of octopus and mammals

(3) Potato and ginger

(4) Potato and sweet potato

81) Which is not true for directional selection ?

(1) Only one extreme value is selected

(2) Mean value never change

(3) Operates in changing environment

(4) Peak shifts in one direction

82) The key concept of Darwinian theory was:

(1) Branching descent

(2) Reproductive isolation

(3) Natural selection

(4) Both (1) & (3)

83) Find out the correct match from the following :-

| | **Column-I** | **Column-II** |
|---|---|---|
| (1) | Jawless fish | 350 bya |
| (2) | Sea weeds | 320 mya |
| (3) | First mammal | Monkey |
| (4) | Coelacanth | First reptile |

(1) 1

(2) 2

(3) 3

(4) 4

84) Match the Column I with Column II and choose the correct option

| Column I | | Column II | |
|---|---|---|---|
| (A) | Neanderthal man existed | (i) | 18,000 years ago, |
| (B) | Agriculture started | (ii) | 1,00,000-40,000 years back |
| (C) | Pre-historic cave art developed | (iii) | About 15 mya |
| (D) | Dryopithecus existed | (iv) | 10,000 years back |

(1) (A) - (i), (B) - (ii), (C) - (iii), (D) - (iv)

(2) (A) - (ii), (B) - (iii), (C) - (i), (D) - (iv)

(3) (A) - (ii), (B) - (iv), (C) - (i), (D) - (iii)

(4) (A) - (i), (B) - (iii), (C) - (iv), (D) - (ii)

85) Which of the following are the example of natural selection due to anthropogenic action?
a. Evolution of antibiotic resistant bacteria
b. Evolution of drug resistant eukaryotic organisms
c. Evolution of different breeds of dogs by intensive breeding programme
d. Darwin’s finches

(1) a, b, d

(2) a, b

(3) a, b, c

(4) a, c

86) Which of the following statement is incorrect?

(1) Different-aged rock sediments contain fossils of different life-forms who probably died during the formation of the particular sediment.

(2) Due to the continental drift pouched mammals of Australia survived because of lack of competition from any other mammal.

(3) Sea weeds and few plants existed probably around 320 mya.

(4) Some of the land reptiles went back into water to evolve into fish like reptiles probably 350 mya.

87) Which of the following statement is correct ?

(1) The rate of disappearance of new forms is linked to the life cycle.

(2) Fitness is the end result of the ability to adapt and get selected by nature.

(3) Branching descent is the key concept of lamarck.

(4) According to Darwin, evolution is driven by use and disuse by organs.

88) Match the following terms with their definitions :

| Term | | Definition | |
|---|---|---|---|
| i. | Homology | A. | Different structures evolving for the same function |
| ii. | Analogy | B. | Same structure evolving in different species for different functions. |
| iii. | Divergent evolution | C. | Selection of similar adaptive features in different organisms. |
| iv. | Convergent evolution | D. | Structures derived from a common ancestor evolving differently |

(1) i-A, ii-B, iii-C, iv-D

(2) i-C, ii-A, iii-B, iv-D

(3) i-D, ii-A, iii-B, iv-C

(4) i-C, ii-D, iii-B, iv-A

89) **Statement-I :** S.L. miller with his experiment proved theory of biogenesis.
**Statement-II :** Theory of biogenesis attempts to explain origin of universe.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

90) A colony of bacteria (say A) growing on a given medium has built in variation in terms of ability to utilise a feed component. A change in the medium composition would bring out only that part of the population (Say B) that can survive under new conditions. This signify :-

(1) In new environment, the population B is more adapted than population A.

(2) In new environment, the population A is more adapted than population B.

(3) In new environment, both the population is equally adapted.

(4) In new environment, both the population will be eliminated.

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 4 | 4 | 1 | 1 | 1 | 3 | 4 | 1 | 4 | 4 | 3 | 4 | 2 | 1 | 1 | 2 | 1 | 3 | 2 | 2 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 4 | 2 | 1 | 3 | 1 | 2 | 1 | 1 | 2 | 3 | 2 | 3 | 1 | 2 | 2 | 1 | 1 | 4 | 3 | 2 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 2 | 1 | 3 | 1 | 4 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 1 | 1 | 2 | 2 | 3 | 3 | 4 | 4 | 2 | 3 | 1 | 4 | 2 | 1 | 4 | 1 | 2 | 2 | 4 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 3 | 2 | 2 | 3 | 2 | 3 | 3 | 3 | 3 | 1 | 4 | 4 | 4 | 3 | 2 | 4 | 1 | 2 | 3 | 4 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 3 | 3 | 1 | 1 | 2 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 1 | 3 | 4 | 4 | 3 | 1 | 4 | 3 | 4 | 2 | 4 | 3 | 4 | 1 | 4 | 3 | 2 | 4 | 2 | 2 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 2 | 3 | 3 | 4 | 2 | 2 | 3 | 1 | 1 | 1 | 3 | 2 | 1 | 2 | 2 | 4 | 2 | 2 | 1 | 2 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 4 | 1 | 1 | 3 | 2 | 1 | 3 | 3 | 1 | 3 | 3 | 1 | 1 | 4 | 2 | 2 | 1 | 3 | 4 | 1 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| A. | 4 | 3 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 4 | 4 | 2 | 1 | 3 | 2 | 3 | 1 | 1 | 3 |
| Q. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | | | | | | | | | | |
| A. | 2 | 4 | 2 | 3 | 2 | 4 | 2 | 3 | 4 | 1 | | | | | | | | | | |

SOLUTIONS

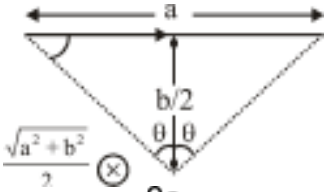
PHYSICS

1) The bar magnet coercivity $4 \times 10^3 Am^{-1}$ i.e., it requires a magnetic intensity $H = 4 \times 10^3 Am^{-1}$ to get demagnetised. Let i be the current carried by solenoid, having n number of turns per metre length, then by definition H = ni. Here $H = 4 \times 10^3$ Amp turn metre$^{-1}$

$$n = \frac{N}{l} = \frac{60}{0.12} = 500 \text{ turn metre}^{-1}$$

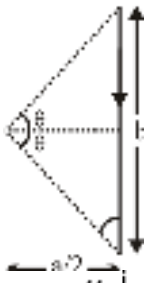
$$\Rightarrow i = \frac{H}{n} = \frac{4 \times 10^3}{500} = 8.0A$$

2) $B_1 = \frac{\mu_0 i}{4\pi\, b/2}\, 2 \sin\theta$




$$= \frac{\mu_0 i}{2\pi b} \frac{2a}{\sqrt{a^2 + b^2}}$$

$$B_1 = \frac{\mu_0 i}{\pi b} \frac{a}{\sqrt{a^2 + b^2}}$$



$$B_2 = \frac{\mu_0 i}{4\pi \frac{a}{2}} \frac{2b}{2\sqrt{a^2 + b^2}} = \frac{\mu_0 i}{\pi a} \frac{b}{\sqrt{a^2 + b^2}}$$

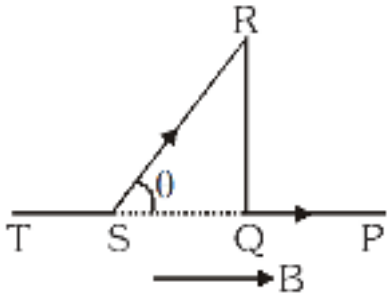
$B_{net} = 2B_1 + 2B_2$

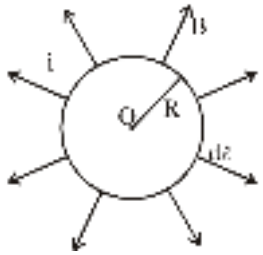
$$= \frac{2\mu_0 i}{\pi \sqrt{a^2 + b^2}} \left[\frac{a}{b} + \frac{b}{a}\right] = \frac{2\mu_0 i}{\pi ab} \sqrt{a^2 + b^2}$$

3) $F = I B \square \sin\theta$

$$= IB(SR) \frac{QR}{SR}$$

$$= IB(QR)$$



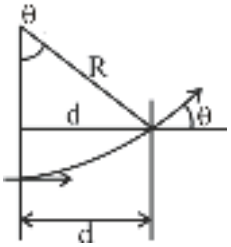




4)

Total force

$F = \oint (i\, d\ell\, B \sin 90^\circ)$

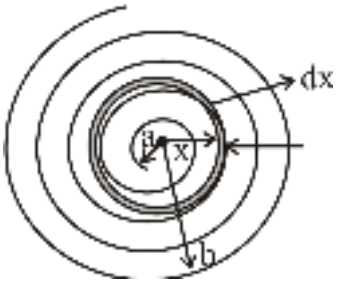
$F = iB \oint d\ell = Bi\,(2\pi R)$




5)

$R = \frac{p}{qB}$

$\sin\theta = \frac{d}{R} = \frac{dqB}{p}$




6)

No. of turns per unit width $= \frac{N}{b-a}$

No. of turns of elementary ring of radius x and thickness dx

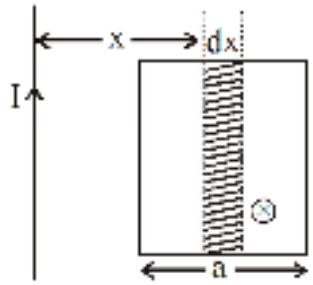
$dN = \frac{Ndx}{(b-a)}$

$\therefore \int dB = \frac{\mu_0 (dN) I}{2x} = \frac{\mu_0 IN}{2(b-a)} \int_a^b \frac{dx}{x}$

$$B = \frac{\mu_0 NI}{2(b-a)} \ell n \left(\frac{b}{a}\right)$$

7)

$$B = \mu_o nI = \mu_o \times \frac{N}{2\pi R} \times I$$
$$= 4\pi \times 10^{-7} \times \frac{500}{2\pi(10 \times 10^{-2})} \times 0.1$$
$$= 10^{-4} T$$




8)

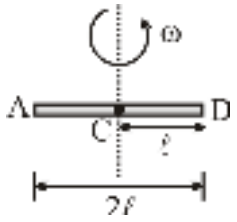
$d\phi = BdA$

$$= \frac{\mu_0 I a dx}{2\pi x}$$

$$\phi = \frac{\mu_0 I a}{2\pi} \int_x^{x+a} \frac{dx}{x}$$

$$MI = \frac{\mu_0 I a}{2\pi} \ln\left(\frac{x+a}{x}\right)$$

$$M = \frac{\mu_0 a}{2\pi} \ln\left(\frac{x+a}{x}\right)$$

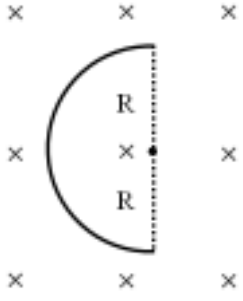



9)

Let centre is C

$V_{CA} = \frac{1}{2} B\omega\ell^2$, $V_{CD} = \frac{1}{2} B\omega\ell^2$,

$V_A = V_D$ So, $V_{AD} = 0$




10)

 e.m.f = BV .....(1)
Here, L = πR

and R = $\frac{L}{\pi}$

e.m.f = BV(2R)

e.m.f = $\frac{2BVL}{\pi}$

11)

Jumps out of the core

12) $e_0$ = ωNBA = (2πv) NBA

= 2 × 3.14 × 100 × 5000 × 0.2 × 0.25 = 157 kV

13) i = $\frac{2\varepsilon}{\sqrt{R^2 + \left[L\omega - \frac{1}{C\omega}\right]^2}}$

Power = $i^2R$ = $\frac{4\varepsilon^2 R}{\left[R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2\right]}$

14)

(A) → Series LCR with resonance or only resistive circuit.
(B) → Phase difference = 90°
 Power factor = 0

15)  $X_C = X_L$ (Resonance)
 $I = \frac{V}{R} = \frac{110}{55} = 2A$

16) Charge on capacitor plates at time t is, q = It.
Electric field between the plates at this instant is

$E = \frac{q}{A\varepsilon_0} = \frac{It}{A\varepsilon_0}$ ..... (i)

Electric flux through the given area $\frac{A}{2}$ is

$\phi_E = \left(\frac{A}{2}\right) E = \frac{It}{2\varepsilon_0}$ Using (i) ..... (ii)

Therefore, displacement current

$I_D = \varepsilon_0 \frac{d\phi_E}{dt} = \varepsilon_0 \frac{d}{dt}\left(\frac{It}{2\varepsilon_0}\right) = \frac{I}{2}$ [Using (ii)]

17) Both magnetic and electric fields have zero average value in a plane electromagnetic

wave.

18) Mirror $M_1$
u = –30 cm, f = –20 cm
v = –60 cm
$m_1 = \frac{-v}{u} = -2$
Mirror $M_2$
u = 20 cm, f = 10 cm
$v = \frac{uf}{u-f} = \frac{20 \times 10}{20-10} = 20cm$
$m_2 = \frac{-v}{u} = -1$
$m_{total} = m_1 m_2 = 2$

19)

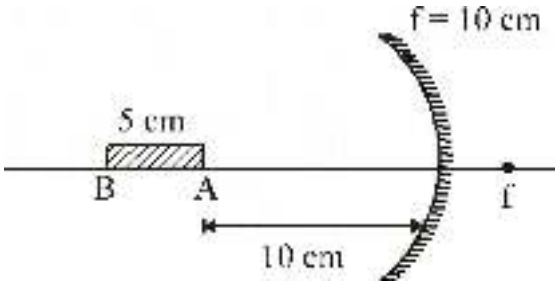



Image of one end of rod will form at infinity
so length of image = ∞

20) $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
$\frac{1}{6} = (1.5-1)\left[\frac{1}{R_1} - \frac{1}{-2R}\right]$
$\frac{1}{3} = \frac{3}{2R}$
R = 4.5 cm

21)

T.I.R. not possible because light ray goes from rarer to denser.

22) For TIR at PQ; θ > C
From geometry of figure θ = 60 i. e. 60 > C
⇒ sin 60° > sin C
$\Rightarrow \frac{\sqrt{3}}{2} > \frac{\mu_{Liquid}}{\mu_{Prism}} \Rightarrow \mu_{Liquid} < \frac{\sqrt{3}}{2} \times \mu_{Prism}$
$\Rightarrow \mu_{Liquid} < \frac{\sqrt{3}}{2} \times 1.5 \Rightarrow \mu_{Liquid} < 1.3$

23)

Initially parallel then diverging.

24) For magnifying glass object should placed in between pole & focus so that virtual, erected and magnified image is formed.

25) $\vec{V}_{\text{image of bird}} - \vec{V}_{\text{fish}} = 9(\hat{i})$

$\vec{V}_{\text{image of bird}} = 9(\hat{i}) - 3(\hat{i}) = 6(\hat{i})$

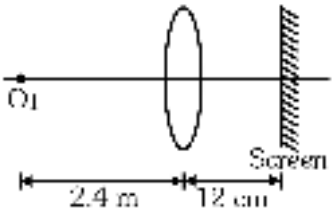
$\vec{V}_{\text{bird}} = \frac{6}{\mu} = \frac{6}{4/3} = 4.5$ m/s

$\vec{V}_{\text{image of fish/bird}} = -\frac{3\hat{i}}{\mu} - 4.5\hat{i}$

$= \frac{-3\hat{i}}{4}(3) - 4.5\hat{i} = -6.75\hat{i}$ m/s.

$V_{\text{image of fish/bird}} = 6.75\ ms^{-1}$.

26)




Applying lens formula

$$\frac{1}{0.12} + \frac{1}{2.4} = \frac{1}{f} \Rightarrow \frac{1}{f} = \frac{210}{24}$$

Upon putting the glass slab, shift of image is

$$\Delta x = t\left(1 - \frac{1}{\mu}\right) = \frac{1}{3}\text{cm}$$

Now $v = 12 - \frac{1}{3} = \frac{35}{3}\text{cm} = \frac{35}{300}\text{m}$

Again apply lens formula

$$\frac{300}{35} + \frac{1}{u} = \frac{1}{f} = \frac{210}{24}$$

Solving u = 5.6 m

Thus shift of object is = 5.6 – 2.4 = 3.2 m

27) $I_{resu} = 4I\cos^2\frac{\pi}{\lambda} \cdot \Delta x$

$$\frac{1}{2} \times 4I = 4I\cos^2\frac{\pi}{\lambda}.\frac{yd}{D}$$

$$y = \frac{\beta}{4}$$

 Distance $= 2y = \frac{\beta}{2}$

28)

$A'^2 = a_1^2 + 2a_1a_2\cos\phi + a_2^2$

$(a')^2 = (3A)^2 + (2A)^2 + 2(3A)(2A)\cos 60°$

$= 9A^2 + 4A^2 + 6A^2 = 19A^2$

29)

Width of central maximum $\propto \frac{\lambda}{a}$
Where:
• λ is the wavelength,
• a is the slit width.

If the slit width a increases, the central maximum will decrease in size.

30) $I = \frac{I_0}{2} \times \cos^2(30^\circ) \times \cos^2(30^\circ)$
$= \frac{I_0}{2} \times \frac{3}{4} \times \frac{3}{4}$
$= \frac{9I_0}{32}$

31)

According to momentum conservation
$m_1v_1 = m_2v_2$
$m_1 2v = m_2 v$
$\frac{m_1}{m_2} = \frac{1}{2}$
$\frac{\frac{4}{3}\pi r_1^3 \rho}{\frac{4}{3}\pi r_2^3 \rho} = \frac{1}{2}$ (m = V$\rho$)
$\frac{r_1}{r_2} = \frac{1}{2^{1/3}}$

32) Due to 10.2 eV photon one photon of energy 10.2 eV will be detected.
Due to 15 eV photon the electron will come out of the atom with energy (15 - 13.6) = 1.4 eV

33) Maximum number of photon is given by all the transitions possible = $\frac{4(4-1)}{2} = 6$
Minimum number of transition = 1,
that is directly jump from 4 to 1.

35)

Photoelectric equation : $eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$.
For wavelength λ : $eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$ (1)

For wavelength 2λ : $e\left(\frac{V}{3}\right) = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0}$ (2)

Subtract (2) from (1) : $eV - \frac{eV}{3} = \frac{hc}{\lambda} - \frac{hc}{2\lambda}$

$\frac{2eV}{3} = \frac{hc}{2\lambda} \Rightarrow eV = \frac{3hc}{4\lambda}$

Substitute in equation (1) : $\frac{3hc}{4\lambda} = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$

$\frac{hc}{\lambda_0} = \frac{hc}{\lambda} - \frac{3hc}{4\lambda} = \frac{hc}{4\lambda}$

$\lambda_0 = 4\lambda$

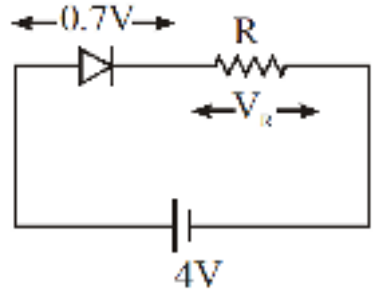
36) $\lambda_{d(neutron)} = \frac{h}{\sqrt{3mkT}}$

$\Rightarrow \lambda_{d(neutron)} \propto \frac{1}{\sqrt{T}}$

$\frac{\lambda_{d(n,\,927^oC)}}{\lambda_{d(n,\,27^oC)}} = \sqrt{\frac{27+273}{927+273}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

$\lambda_{d(n,\,927^oC)} = \frac{\lambda_{d(n,27^oC)}}{2} = \frac{\lambda}{2}$

37) By drawing circuit and using KVL, we have




4V = 0.7V + VR

= 0.7V + IR

⇒ IR = 4 – 0.7 = 3.3 volt

$\Rightarrow R_{min} = \frac{3.3\,volt}{I_{max}} = \frac{3.3\,volt}{5\,mA} = \frac{3.3V}{5 \times 10^{-3}A} = 660\ \Omega$

38)

Here diode is R.B.

So reading of $A_1 = 0$

Reading of $A_2 = \frac{2}{10} = 0.2$ A

39)

$\Delta E_g$ decreases with increasing temperature.

40) $\overline{\overline{A+B}} = A + B$ OR gate

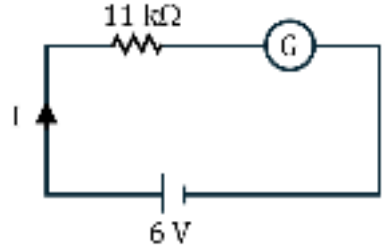
$\overline{A.B} = A . B$ AND gate

$\bar{A} + \bar{B} = \overline{A.B}$ NAND gate

$\bar{A} . \bar{B} = \overline{A+B}$ NOR gate

41)

If no shunt is present,

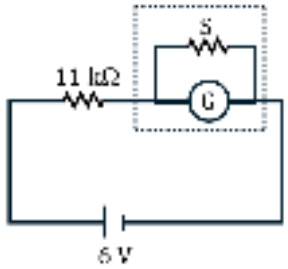



$I = 9 \times 60\ \mu A = 540 \times 10^{-6}\ A$

$$\frac{6}{11000+G} = 540 \times 10^{-6}$$

$$G = \frac{1000}{9}$$

If shunt is applied :




From the standard relation from half deflection method.

$$G = \frac{RS}{R-S} \Rightarrow \frac{1000}{9} = \frac{11000\,S}{11000-S}$$

S = 110 Ω

42)

This question is based on I–V Characteristics of a P–N Junction Diode (Static Resistance)

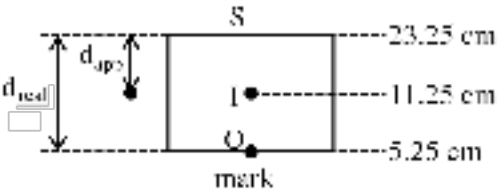
(1) At V = – 8V

$$R = \frac{-8}{-1\mu} = 8M\Omega$$

(2) At $I_d$ = 10 mA

$$R = \frac{\Delta V}{\Delta I} = \frac{0.8-0.6}{(15-5)\times 10^{-3}} = 20\Omega$$

43)




Real depth = (23.25 – 5.25) cm = 18.00 cm

Apparent depth
= (23.25 – 11.25) cm = 12.00 cm

$$RI = \frac{\text{Real depth}}{\text{App. depth}} = \frac{18.00}{12.00} = 1.5$$

44) Maximum safe zener current

$$I_{z(max)} = \frac{P_z}{V_z} = \frac{1.6}{8} = 0.2A$$

Load current

$$I_L = \frac{V_z}{R_L} = \frac{8}{10} = 0.8A$$

Worst-case condition

Worst case of zener safety occurs at maximum input voltage (10 V).

Total current through series resistance:

$I_S = I_L + I_{Z(max)} = 0.8 + 0.2 = 1.0$ A

Voltage across series resistance:

$V_R = V_{max} - V_Z = 10 - 8 = 2$ V

$$R_S = \frac{V_R}{I_S} = \frac{2}{1.0} = 2\Omega$$

45)

Mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{-1}{u} + \frac{1}{f}$$

$$\Rightarrow y = mx + c$$

Slope m = –1

Intercept $c = \frac{1}{f} \Rightarrow f = \frac{1}{c}$

$$f = \frac{1}{-0.10} = -10 \text{ cm}$$

CHEMISTRY

46) Due to synergic bonding

$C–O_{B.L} \propto$ –ve charge

47)

Crystalline form of boron is unreactive.

48)

$Pt^{+2}$ ⇒ C.No. = 4
$Pt^{+4}$ ⇒ C.No. = 6

49)

Cobalt(III) → Oxidation state = +3

Ligands:

A. $2Cl^-$ → Total charge = –2
B. 2 en (neutral) → Total charge = 0

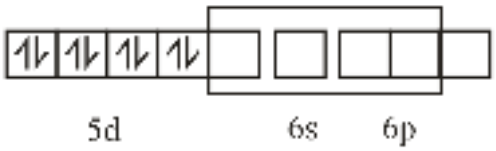
Overall charge = +3 (Co) + (–2 from $Cl^-$) = +1
So, correct formula: $[CoCl_2(en)_2]^+$

50)

$[NiCl_4]^{2-}$
$Ni^{+2} = 3d^8\ 4s^0$
$Cl^-$ WFL so no pairing of $e^-$

51)

$[PtCl_4]^{2-}$ involves $dsp^2$ hybridization with orbitals of different n values.
$[PtCl_4]^{2-}$
$Pt^{+2} = 5d^8$ =




52)

Fact/Observation
NCERT Lab Manual

53)

**Concept:** Ligand field strength (Δ) ∝ energy of absorption (E) ∝1/λ
Arrange ligands by field strength (strong → weak):
$CN^- > NH_3 > NO_2^- > Cl^-$
$[FeCl_6]^{3-}$ has $Cl^-$ (weak-field) → highest wavelength of absorption.

54) $d^6$-low spin → $t^6_{2g}\ e_g^0$
CFSE = $[-0.4n_{2g} + 0.6\ n_{eg}]\Delta_0 + nP$
= $[-0.4 \times 6 + 0.6 \times 0]\Delta_o + 3P$

$= -\frac{12}{5}\Delta^0 + 3P$

55)

As per structure
(I) $[Ma_3b_3]$ type
(II) $[M(AA)_3]$

56)

As per concept.

57)

Reason is correct explanation of the assertion because identical position means no possibility of different spatial arrangement leading to G.I.

58)

$[Fe(en)_2(H_2O)_2]^{+2} + en \rightarrow [Fe(en)_3]^{+2}$
$Fe^{+2} = 3d^6$, $t_{2g}^6\ eg^0$ low spin, diamagnetic

59) Trans isomer

60) $[Rh(PPh_3)_3\ Cl]$ : Wilkinson catalyst
Every ligand act as a strong field ligand with 4d transition series element
So [C.N. =4] → $dsp^2$ square planar

61)

Highest fluoride of Mn is $MnF_4$

62)

In acidic medium = $Cr_2O_7^{-2}$ exists, but in basic medium $CrO_4^{-2}$ will exists.
→ Ce, Tb can shown +3, +4 O.S.
→ $KMnO_4$ is violet in colour due to charge transfer spectrum.
→ AgCl is colourless (due to polarisation),
$CrO_2Cl_2$ → is Red orange coloured vapour due charge transfer spectra

63)

Boiling point :
$BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

64) The size of $Ln^{3+}$ ions decreases gradually as the atomic number increases (La→Lu) due to Lanthanoid Contraction. The graph showing a steady decline in radius (Y-axis) with increasing atomic number (X-axis) is the correct one.

65)

$NiCl_2$ solution → Green

66)

NCERT-XII Pg.# 201

67)

As per concept.

68)

$FeCl_3 + 3KSCN \rightarrow Fe(SCN)_3$
Blood red colour

69) +R effect groups will decrease carbanion stability.

70)

Rate in $E_2 \propto$ stablity of alkene

71)

Ozonolysis

72)

Haloform

73)

Mixed of $E_1$, $E_2$, SN, $SN^2$

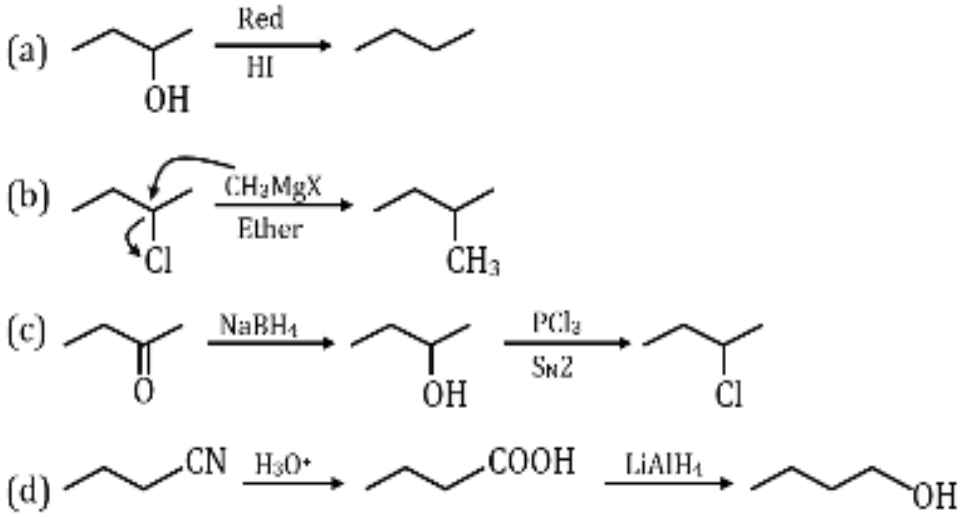
74)

**Solution :-**
**(P) $CH_3$–C ≡ CH + dil. $H_2SO_4$ + $Hg^{2+}$**. This is **keto-enol tautomerism** via **acid-catalyzed hydration of alkyne**, where the terminal alkyne forms a methyl ketone. Product :
$CH_3$–CO–$CH_3$ → (d)

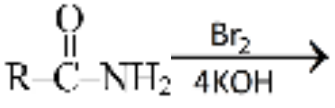
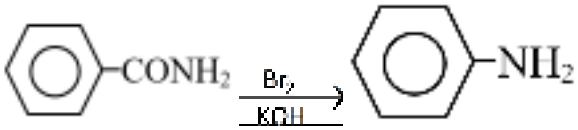
**(Q) m-nitrotoluene + $CrO_2Cl_2$ (Etard reaction)** $CrO_2Cl_2$ oxidizes the benzylic $-CH_3$ to $-CHO$ in the presence of $NO_2$, forming **m-nitrobenzaldehyde.** Product : CHO attached to $NO_2$–substituted benzene → (c)
**(R) $CH_3$–C ≡ N + PhMgBr → hydrolysis** Grignard reagent adds to nitrile forming an imine intermediate, which upon hydrolysis gives a ketone with phenyl and methyl groups : Ph–CO–$CH_3$ → (b)
**(S) Benzene + EtCOCl + $AlCl_3$ (in $CS_2$)** Classic **Friedel-Craft acylation,** forming benzophenone-type ketone, where ethyl group adds as an acyl. product : Benzene ring with CO–Et → (a)
**Final Answer :-** (1)

75)

**Solution :**




The correct match is a → s, b → q, c → r, d → p
**Final answer :** 1

76) **Solution :**  $R–NH_2 + K_2CO_3 + KBr + H_2O$
(1) Number of carbon atom in product amine is one less than amide used as reactant ⇒ correct
(2) R–CO–NHBr and R–N=C=O are formed as intermediate ⇒ Corect
(3) 4 moles of KOH consumed in reaction ⇒ correct
(4) aniline 
**Final Answer :** Option (4)

77)

Lassigne extract $\xrightarrow{FeCl_3}$ N, S
Blood red

78)

Fact, NCERT Based

79) **Problem Statement:** The question asks which of the given sugar molecules is a non-reducing sugar. Non-reducing sugars do not have a free aldehyde or ketone group and therefore do not react in typical reduction tests.
**Underlying Concept:** Sugars can be classified as reducing or non-reducing based on the presence of a free aldehyde or ketone group. Reducing sugars have a free anomeric carbon capable of open-chain form, allowing them to reduce mild oxidizing agents. Non-reducing sugars do not have this free reactive group, often due to glycosidic linkage involving the anomeric carbons of both monosaccharide units.
**Tips and Tricks:** Remember that sucrose is a common example of a non-reducing sugar because its glycosidic bond links the anomeric carbons of glucose and fructose, preventing free aldehyde or ketone groups.
**Common Mistakes:** Mistaking all disaccharides as reducing sugars. Some disaccharides like sucrose are non-reducing because of their linkage.
**Why Other Options Are Incorrect?:** Option (a) and (d) are reducing sugars because they have free anomeric carbons available. For example, maltose (a) and lactose (d) have a free aldehyde group in one monosaccharide unit allowing reduction.

80)

Fact (NCERT)

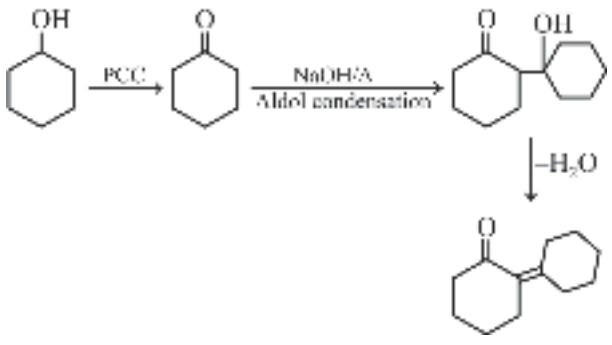
81)

Amine (NCERT)

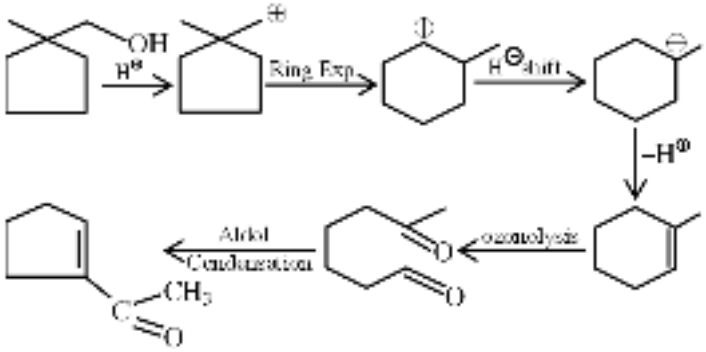
82)

Boiling point increases on increasing number of carbon.

83)



84)

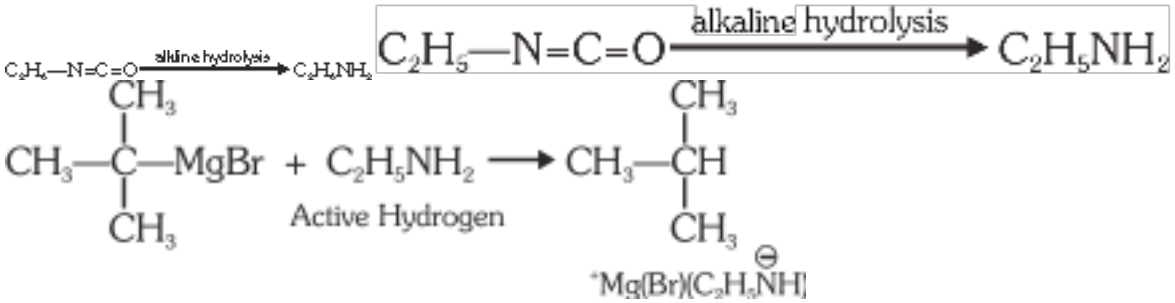



85)

Rate of Hydrolysis ∝ EWG

86) **Solution :-** Carbonyl compound having no. α-H give cannizzaro reaction. Two molecule participate in the reaction in which one get oxidized to acid (salt) and other gets reduced to alcohol.
if two different molecule participate in reaction then more reactive molecules gets oxidised to acid and less reactive molecule gets reduced to alcohol.
In this case HCHO is more reactive and Ph-CHO is less reactive towards cannizzaro reaction so, oxidation of HCHO and Reduction of Ph-CHO takes Place.
Therefore B and C process are involved in the reaction.
**Final Answer :-** (3)

87) **Solution :**




• Hydrolysis : $C_2H_5$–NCO + $OH^-/H_2O$ → $C_2H_5$ (ethylamine) + $CO_2$.
• Grignard quench : forms t-Bu-H (isobutane) and MgBr amide.
**Final answer :** 3

88)

**Solution :**
Hinsberg's reagent

| **Types of amine** | **Reaction with Hinsberg's reagent ($C_6H_5SO_2Cl$)** | **Product Solubility in Alkali** |
|---|---|---|
| Primary (1°) | Forms sulfonamide with acidic H | Soluble (forms salt) |

| | | |
|---|---|---|
| Secondary (2°) | Forms sulfonamide without acidic H | Insoluble (no salt) |
| Tertiary (3°) | Does not react | — |

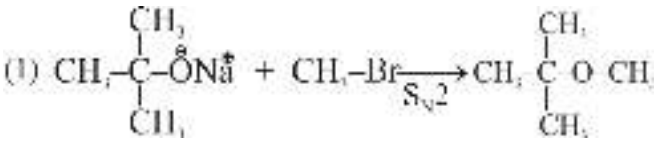
 Differentiates 1°, 2°, and 3° amines.

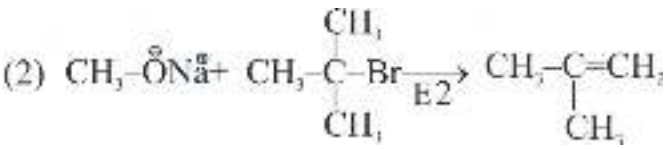
| **Given compounds** | **Functional group** | **Hinsberg's Test** |
|---|---|---|
| 1 $CH_3\ CH_2\ CH_2$ – $NH_2$ | 1° amines |  |
| 2 $CH_3CH_2$–NH–$CH_3$ | 2° amines |  |
| 3 $CH_3CH_2$–OH | Alcohol |  |
| 4 $(CH_3)_3N$ | 3° amines |  |

Compound ($CH_3\ CH_2\ CH_2$ – $NH_2$) that will react with Hinsberg's reagent to give a solid which dissolves in alkali.
**Final answer :** option (1)

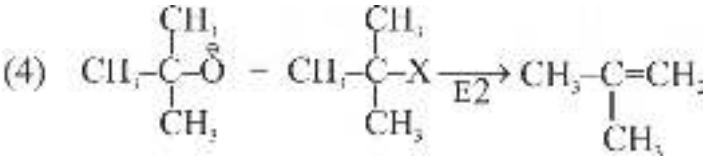
89)

**Solution:**



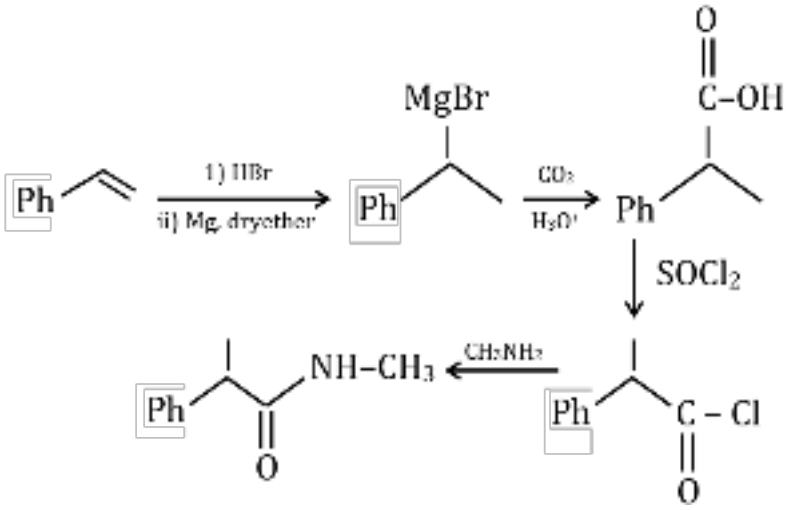




**Final Answer:** Option (1)

90) **Concept:**




BIOLOGY

91) NCERT Pg.#61, 63, 95

92) NCERT Pg. # 60, 61

93) NCERT-XII, Pg. 98, 99

94) NCERT (XII) Pg. # 96, 98

95)

NCERT-XII, Pg. 104

96)

NCERT-XII, Pg. # 90

97) NCERT-XII, Pg. # 81

98) NCERT XII Pg # 96

99) NCERT (XII) Pg. # 81

100) NCERT-XII, Pg. # 101

101) NCERT-XII, Pg. # 93, 98

102)

NCERT-XII Pg. # 90

103)

NCERT-XII, Pg # 105

104)

NCERT-XII, Pg # 83

105) NCERT-XII, Pg. # 60

106) NCERT-XII Pg. # 75

107) NCERT-XII, Pg. # 58, 60, 63

108) NCERT-XII, Pg. # 74

109) NCERT-XII, Pg. # 59

110) NCERT-XII, Pg. # 69

111) NCERT-XII, Pg. # 68

112) NCERT-XII, Pg. # 73

113) NCERT-XII, Pg. # 76

114)

NCERT-XII, Pg # 197, 200

115)

NCERT-XII, Pg. # 200

116) NCERT XII Page # 198

117) NCERT XII Pg # 197-202

118)

NCERT-XII, Pg # 197-202
Hint : Dog-Ticks, Human Liver Fluke-Snail, Copepods-Fish

119)

NCERT-XII, Pg. # 193

120)

NCERT-XII, Pg. # 195

121) NCERT-XII, Pg # 191, 192

122)

NCERT-XII, Pg.# 194

123)

NCERT-XII, Pg. # 213

124)

NCERT-XII, Pg. # 214

125)

NCERT-XII, Pg. # 213

126) NCERT-XII, Pg # 214

127)

NCERT-XII, Pg. 207, 208

128) NCERT-XII, Pg. # 211

129) NCERT XII Pg. # 207

130)

NCERT-XII, Pg. # 218

131)

NCERT-XII, Pg. # 216, 217

132) NCERT XII, Pg. # 222

133)

NCERT-XII, Pg. # 222, 223

134)

NCERT-XII, Pg. # 224, 225

135) NCERT-XII, Pg. # 225

136) NCERT-XII, Pg # 170

137) NCERT-XII, Pg # 166

138) NCERT-XII, Pg # 169

139) NCERT-XII, Pg # 168, 169

140) NCERT-XII, Pg # 171

141) NCERT-XII, Pg. # 168

142) NCERT-XII, Pg # 169

143) NCERT-XII, Pg # 181

144) NCERT-XII, Pg # 180, 181, 182, 183

145)

NCERT-XII, Pg # 185
**Correct Answer :** Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

146) NCERT-XII, Pg. # 179, 180, 182, 183, 184

147)

NCERT-XII, Pg # 181, 185

148)

NCERT-XII, Pg # 180, 182, 185

149) NCERT-XII Pg. # 177

150)

NCERT-XII, Pg # 183

151)

NCERT-XII, Pg # 168, 169, 170, 175

152) NCERT-XII, Pg.# 179,180,182, 172

153) NCERT-XII, Pg # 157

154) NCERT-XII Pg. # 152, 153, 156, 157

155) NCERT-XII, Pg # 151, 153

156) NCERT-XII, Pg. # 151

157) NCERT-XII, Pg. # 153

158)

NCERT-XII, Pg # 153

159)

NCERT-XII, Pg # 147

160)

NCERT-XII, Pg # 157

161)

NCERT-XII, Pg # 146

162)

NCERT-XII, Pg # 156

163)

NCERT-XII, Pg # 155

164) NCERT XII Pg. No. # 142, 143

165) NCERT, Pg. # 140

166) NCERT Pg#131

167) NCERT, Pg. # 131

168) NCERT, Pg. # 131, 133, 138

169) NCERT-XII Pg. # 142-143 (Para-7.5)

170)

NCERT-XII, Pg # 130

171)

NCERT-XII, Pg # 136

172)

NCERT-XII, Pg. # 129

173)

NCERT-XII, Pg # 138

174) NCERT Pg # 124, 125

175) NCERT Pg # 115, 116

176) NCERT Pg # 113, 122, 124

177) NCERT, Pg. # 118

178) NCERT Pg. # 115

179) NCERT-Pg. # 111

180)

NCERT-XII, Pg. # 140